

Places and People Under Heterogeneous Federal Reserve Policy

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Abstract. I study the *places* (local labor markets) and *people* (workers) consequences of the Federal Reserve's heterogeneous policy response to the bank failures that occurred in Mississippi at the onset of the Great Depression. During this period, the St. Louis district contracted its credit supply while the Atlanta district did the opposite—a rare natural experiment in monetary policy implementation. By using a Regression Discontinuity design and randomization inference, I show that (1) The bank failure rate was significantly smaller in the Atlanta district relative to the St. Louis district. (2) There were economically and statistically significant long-term differences in labor market outcomes and migration. Ten years after the initial shock, residents of the Atlanta district side of Mississippi (place) were less likely to be unemployed, less likely to be in work relief, less likely to be in the labor force, and had higher weekly income relative to their St. Louis counterparts. (3) By tracking individuals who were originally near the boundary in 1930, I find that ten years later, individuals originally on the St. Louis side were more likely to migrate. Importantly, for these individuals (people), most of the differences relative to their Atlanta-side counterparts were mitigated, suggesting that migration served as a channel through which individuals could counteract the negative shock faced by places on the St. Louis-side.

Keywords: Monetary Policy, Labor Markets, Places versus People, Internal Migration, Great Depression, Bank Failures.

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Introduction

In 1930, heterogeneous policy approaches by the St. Louis and Atlanta districts of the Federal Reserve led to one of the best natural experiments we have on unconventional monetary policy implementation. Existing studies have documented some short- and medium-run effects of these policies ([Richardson and Troost \(2009\)](#); [Ziebarth \(2013\)](#); [Hansen and Ziebarth \(2017\)](#); [Jalil \(2014\)](#)). However, due largely to data and methodological limitations, the literature has overlooked whether these policies affected *places* (i.e. local labor markets) and *people* (i.e. workers), especially in the long run.¹ I tackle these questions by leveraging granular data and improved econometric techniques, and uncover three key findings: monetary policy has long-run effects on local labor markets, monetary policy has much smaller long-run effects on people, and these two results can be reconciled by the response of migration.

Yet estimating such effects is challenging for several reasons. First, at present there is no monetary policy variation in *implementation* at sub-national levels. Second, even if such variation existed, we still face the fundamental issue that monetary policy is endogenous, which is key for identification of causal effects. Third, we would need to have granular data on workers' outcomes and locations, for at least two periods—before and after the policy implementation—in order to attribute any observed effects to the policy itself. In this paper I construct a setting to overcome those challenges.

To tackle the challenges, I first exploit the early institutional structure of the Federal Reserve System. In its youth, policy implementation exhibited variation at the Federal Reserve District level. I exploit this historical feature by focusing on the policy divergence between the St. Louis and Atlanta districts in the aftermath of the collapse of Caldwell and Company—the largest financial conglomerate in

¹Here I use the terms *places* and *people* to refer to local labor markets and individual workers in the same spirit as [Autor et al. \(2025\)](#), even though they use them in a different, unrelated, context.

the South at the time. Following the wave of bank failures in late 1930 and early 1931, the St. Louis district responded by contracting credit its credit supply, while the Atlanta district expanded it. Building on [Richardson and Troost \(2009\)](#), I treat this as a natural quasi-experiment of monetary intervention.

Second, I leverage modern Regression Discontinuity techniques applied to this setting. I generate a data-driven, optimal narrow window around the districts boundary in Mississippi in such a way that observed covariates are well balanced and, thus, the only meaningful variation comes from the policy regime to which either side is exposed. I validate this identifying assumption using both bank-level data manually transcribed from *Rand McNally Bankers' Directories* and the full-count 1930 Census. I use these to demonstrate that prior to the collapse, banking conditions, labor market characteristics, and other key covariates are indistinguishable on both sides of the boundary. Thus concluding that we cannot reject the assumption that treatment assignment (i.e., exposure to the Atlanta district policy) is as-if random.

Third, I utilize granular data. On the one hand, from the place perspective, I use the full-count Decennial Censuses of 1930 and 1940, and build upon [Berkes et al. \(2023\)](#) to infer the location of nearly 75% of Mississippi residents, which allows me to estimate place effects ten years later. On the other hand, from the people perspective, I follow individuals between censuses by applying the linkage methodology of [Abramitzky et al. \(2021\)](#), which enables the estimation of long-run effects at the individual level.

Thus by combining the historical quasi-experiment, a data-driven Regression Discontinuity design, and detailed microdata, I am able to study both the short-term effects on the banking sector and the long-term effects on labor market outcomes from both the place and people perspective.

I find that in the very short term, within two months of the initial shock, the banks located in the Atlanta side of the boundary were less likely to fail *permanently*

by around 14 percentage points, relative to banks on the St. Louis side. This result is both economically and statistically significant as it exceeds the average failure for all banks near the boundary by 4 percentage points. This result is roughly in line with those found by both [Richardson and Troost \(2009\)](#) and [Jalil \(1929\)](#), who used different methodologies and data.

I now turn to the results on local labor markets, that is, the effects on residents of Mississippi ten years later. I find that ten years after the initial shock, the probability of being unemployed was 2 percentage points lower for residents of the Atlanta side of the boundary. If we exclude Public Emergency Work from the definition of unemployment, then the number is still 1 percentage point lower. Both of these results are statistically and economically significant, as they represent about half of the mean (respectively). On the other hand, I don't find discernible differences in the probability of being employed. These results combined imply a strong negative effect on labor force participation, indeed Atlanta residents were 3 percentage points less likely to participate in the labor force than the St Louis residents, and this number is both economically and statistically significant as well. This seemingly puzzling result, is in line line with recent literature that finds an increase in labor force participation after a contractionary monetary policy shock ([Graves et al. \(2023\)](#)).

When it comes to local labor markets compensation and the intensive margin, I find no significant differences in terms of hours worked the week before the Decennial Census was taken. Similarly, I do not find significant differences in yearly income earned in 1939. However, there is an economically and statistically significant difference in weeks worked the previous year, the Atlanta residents had to work around two less weeks on average. Since the mean weeks worked was 24, the estimate represents around 8 percent of the mean. This result suggests that the weekly wages for Atlanta district residents were significantly higher than for the St. Louis district residents.

Turning now to the effects on workers, meaning, people who lived in Mississippi in 1930 near the boundary between districts and that in 1940 were living anywhere in the country, I find a few important results. First, almost all significant differences found on places are absent in people. Second, there is still a significant difference present in the probability of being unemployed. Workers originally in the Atlanta district were 1 percentage point less likely to be unemployed in 1940 regardless of their new residence. Importantly, even though the point estimate is not far from the one found on places, it represents around 14% of the mean. Still economically significant, but about a third of the size found when comparing places. Third, even though most results on workers are null, I show that they are still significantly different from the results on places. Furthermore, the sign of this comparison essentially always goes in opposite direction than the sign on places, which suggests people were likely actively counteracting the negative results of the shock.

A potential channel that can reconcile the above results is migration. Indeed I find that people who were originally residents of the St. Louis district near the boundary in 1930, were 5 percentage points more likely to out-migrate out of their counties of origin than the originally residents of the Atlanta district. This number is 2 percentage points when using out of state migration instead. Regardless of whether I use out of county or out of state measures of internal migration, the results are highly significant both statistically and economically as they represent about 10 to 11 percent of the mean, respectively. Taking together, these results strongly suggest that people actively choose to mitigate the differentially negative shock that St. Louis district places received at the end of 1930.

This paper is related to several strands of the literature. First, it relates to the literature that attempts to identify the effects of monetary interventions on bank failures and other economic outcomes (e.g. [Richardson and Troost \(2009\)](#), [Ziebarth \(2013\)](#), [Jalil \(2014\)](#)). Relative to this literature, this paper contributes in a number of ways. First, by using the full count Decennial Census and sub-

county locations I'm able to study the effects on local labor markets as a whole. Second, I use a data-driven optimally chosen window around the federal reserve districts boundary which helps in avoiding the risk of cherry picking. By matching individuals across decennial censuses, I'm able to estimate long-term effects of such monetary interventions. Finally, I study place effects, people effects, and the role of migration which was absent from this literature until now.

The paper also relates to the long-standing comparisons between people versus place effects of policies. Some recent examples in the literature studied carbon emissions ([Lyubich \(2025\)](#)), weather and mortality ([Deryugina and Molitor \(2020\)](#)), the China shock ([Autor et al. \(2025\)](#)), among others.² Relative to this literature, I study the effects of a monetary policy shock, this is the first paper that does this since monetary policy currently is implemented at the national level.

There is an ample literature on the effects of monetary policy on labor market outcomes, to name just a few [Romer and Romer \(2023\)](#), [Graves et al. \(2023\)](#), and recently, [Coglianese et al. \(2025\)](#), [ERCEG and LEVIN \(2014\)](#), [Dolado et al. \(2021\)](#). Relative to this literature I provide new evidence. First, I study individual level (and georeferenced where applicable) data which is not possible under typical empirical strategies in macroeconomics. I also find effects on migration, which are novel relative to this literature. In addition I show that monetary policy can have long term consequences, as I find effects for at least 10 years. The only other paper that studies long term effects is [Jordà et al. \(2024\)](#) and does so with aggregate data.

My paper also relates with the burgeoning literature on the causes of internal migration. Indeed, the literature has shown that weather conditions ([Deryugina and Molitor \(2020\)](#), [Baerlocher et al. \(2025\)](#)), political decisions [Cortes and Sant'Anna \(2024\)](#) , in-migration ([Boustan et al. \(2010\)](#)), labor demand from other places ([Howard \(2020\)](#)), are among some of the principal reasons to out-migrate. Relative

²See [Chyn and Katz \(2021\)](#), [Neumark and Simpson \(2015\)](#), [Kline and Moretti \(2014\)](#) for recent reviews of the literature.

to this literature, this is the first paper to show a link between monetary policy decisions and out-migration.

Finally, my paper also relates to the rapidly increasing literature on Regression Discontinuity (RD) Designs, in particular those that utilize Randomization Inference which was popularized by [Cattaneo et al. \(2015\)](#) and [Cattaneo et al. \(2016\)](#).³ Relative to this literature, and building on ideas from the cluster-robust literature (e.g. [Barrios et al. \(2012\)](#) and [Bugni et al. \(2025\)](#)) I show via Monte Carlo simulations that the base-line statistic used in the conventional (non-cluster) inference does not achieve correct size and then I propose to use a cluster-robust studentized-t statistic which does achieve correct size.

The rest of the paper is structured as follows, first in the next section I describe the historical context while discussing the natural quasi-experiment in some extent. Then, I talk about the data and methods used in sections II and III respectively. Later I show the results and provide some discussion in section IV. Finally I conclude with some closing remarks in the last section of the paper.

1 Historical Background

1.1 The Federal Reserve Policy during its early years

In its youth, the Federal Reserve System's policy had two crucial features that it no longer has. The first was heterogeneity in the *implementation* of policy, as each Federal Reserve District decided its own policies. The second was that the main policy instruments were not the federal funds rate but the discount rate, the discounting of bills, and the purchase and sale of government securities. These two characteristics shaped the policy behavior of different districts during the Great Depression, particularly in reaction to widespread bank failures that occurred

³See [Lee and Lemieux \(2010\)](#), [Ritzwoller et al. \(2025\)](#), and [Cattaneo and Titiunik \(2022\)](#) for reviews on RD designs more generally.

at the time. One of the key events that caused a wave of bank failures was the collapse of Caldwell & Company.

1.2 The importance of Caldwell & Co. and its collapse

Caldwell & Co. was a fast-growing conglomerate established in 1917 as a municipal bonds house in Nashville, Tennessee.⁴ Headed by Rogers Clarke Caldwell, within the span of thirteen years the firm came to dominate the financial scene in the South and Caldwell himself got to be known as the “Morgan of the South”. By the end of 1929, the conglomerate managed insurance companies, banks, industrial companies, newspapers, and investment trusts, controlling almost 500 million in assets—equivalent to about 83% of the total assets of the banks in Tennessee and roughly 174% of those in Mississippi. This expansion, however, was funded primarily by non-liquid, borrowed funds. By the mid of 1930, its liquid reserves represented a mere 4% of its demand liabilities. In practice this meant that Caldwell and Co. had a tiny reserve of cash and liquid assets to meet withdrawals. Soon enough, rumors of its bad financial state spread and eventually collapsed.

The collapse began with the failure of the Bank of Tennessee —the company’s most important bank—on Friday, November 7, 1930, which was publicly announced the following morning. It culminated just days later when Caldwell and Company itself collapsed on November 13, with the announcement made on the 14th. The fall of Caldwell and Co. had a tremendous impact on the Southern banking system. Within a couple of months, [Mitchener and Richardson \(2019\)](#) document that there were 756 bank suspensions associated with this event. I show the permanent bank suspensions related to the Caldwell collapse in Figure (8). These suspensions were heavily concentrated in the Southern states, Mississippi among them. I show in Figure (1) that very quickly after the mentioned collapse, the bank failures in the

⁴The most detailed account of the Caldwell and Company conglomerate can be found in [McFerrin \(1969\)](#).

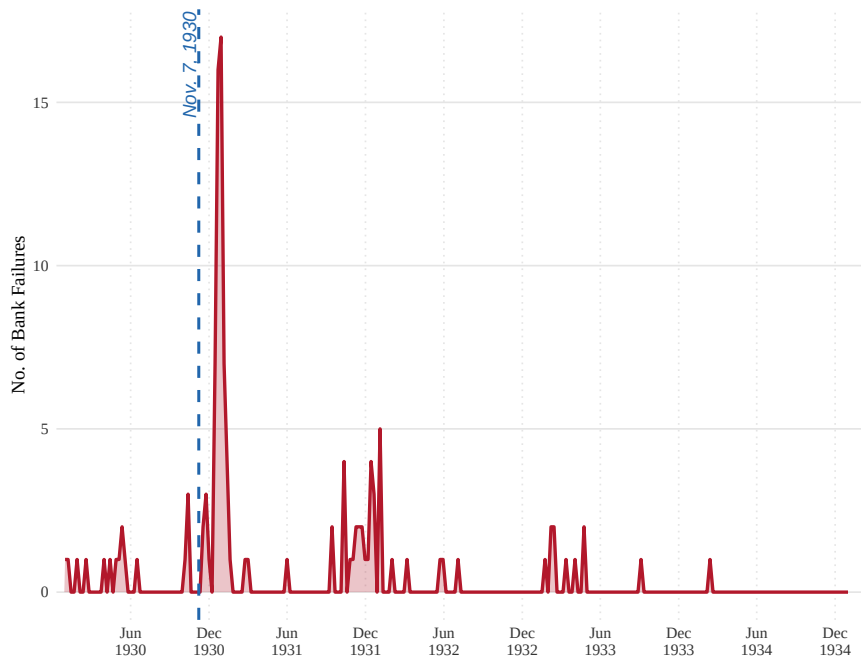


Figure 1. Weekly Bank Failures in Mississippi. Source: Rand McNally Banker's Directory, several years.

state went up to unprecedented levels, and was the only time during the Great Depression that a panic of such level happened in the state.

1.3 Heterogeneous policy response as a Natural Experiment

Among the affected states, Mississippi offers an ideal setting to study monetary policy responses to the failures, as the policy responses of its Federal Reserve Districts are a good example of a natural experiment ([Richardson and Troost \(2009\)](#), [Ziebarth \(2013\)](#), [Hansen and Ziebarth \(2017\)](#)). There are several reasons why. First, the state is split in two different Federal Reserve districts, allowing one to control for any state-level policies implemented during the Great Depression. At the same time, Mississippi the state was fully contained within other relevant banking-related boundaries ([Richardson and Troost \(2009\)](#)). Second, Mississippi geographically distant from reserve cities and does not contain any Federal Reserve branches in it, making it unlikely that district-level policy decisions were influenced by local conditions idiosyncratic to the state. Third, economic

characteristics near the district boundary were balanced across sides. While previous work has documented this to some extent, I go further in the methodology section by estimating the optimal window around the boundary where covariate balance holds.

Finally—and crucially—the two different reserve districts were not just physically separate but they also implemented fundamentally different policy regimes. The St. Louis District was a strong proponent of the Real Bills doctrine. In essence, this doctrine, postulates that the central bank’s responsibility was to extend credit that was backed up only by productive purposes. This view is inherently pro-cyclical because when the economy is collapsing, people would likely demand more credit, but this credit will be denied because it will not be justified by a productive endeavor. In contrast, the Atlanta District held a much more interventionist approach, consistent with the Bagehot’s rule, which in essence promoted central banks to act as lenders of last resort in times of panics.

The contrasting policy regimes of the two districts were clearly reflected in their actual policy responses to the failures triggered by the Caldwell & Co. collapse. In Figure (2) it can be seen that shortly after the collapse of Caldwell and Company, the total credit supplied by the Atlanta Fed increased about 50 percent relative to the St. Louis Fed credit provision. In Figure (11) I show the decomposition of this credit provision which disentangle a few facts. First—and crucially—the Atlanta district *increased* total credit provision while the St. Louis *decreased* it during the same period. Second, most of the contraction for St. Louis comes from the selling of government securities. Third, most of the increase in the Atlanta credit provision comes from the discount window, indicating that the Atlanta fed was aggressive in providing short term credit to the banking system. This intervention helped in mitigated banking panics during the great depression (Richardson and Troost (2009)). See for example the suggestive evidence in Figure (14); I test this more formally in the results section.

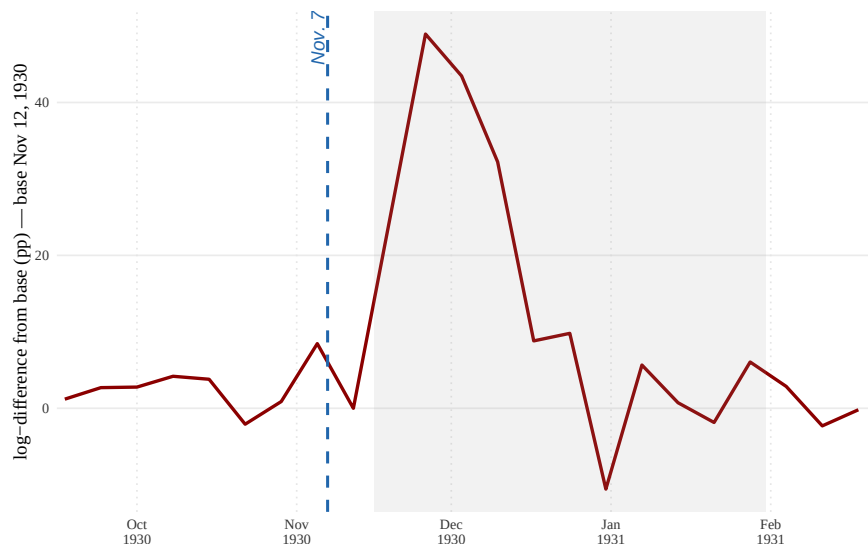


Figure 2. Difference in Total Credit Supplied Atlanta - St. Louis (%). This graph shows $D_t = 100 \times \left(\ln \frac{A_t}{S_t} - \ln \frac{A_{t_0}}{S_{t_0}} \right)$, the weekly log difference between the credit supplied by the Atlanta minus the St. Louis district relative to Nov 12, 1930 levels. Values are expressed in percentage points. The shaded area is the Caldwell-related period. Sources: Federal Reserve Bulletin and Annual Report, several years.

The take away from this part is thus that we have a natural experiment in monetary intervention which featured a counter-cyclical and highly active response by the Atlanta Fed, and a contractionary, pro-cyclical response by the St. Louis Fed. I exploit this heterogeneity to study both the very short-run effects on the banking sector and the long-run effects on places and people.

2 Data

My two main data sources are bank-level data that I collect from the the Rand McNally Directories and individual-level data from the Decennial Censuses. I describe them now.

2.1 Bank-level data

The Rand McNally Bankers Directory, or the “Blue Book”, was a regular semi-annual banking-oriented publication that contained information about essentially

every bank in the US. This source of information had been increasingly used in the literature, although there is still no systematic collection of all of them, to the best of my knowledge.⁵

The data scattered in those directories contains, among other things: the name and type of each bank (state, national, or private); the name of its president; a breakdown of liabilities and resources (typically reported as of the end of the month prior to the directory's publication); the names of principal correspondents; and, crucially, the town and county where each bank was located. A typical page in the directories can be seen in Figure 15.

I manually extract the location details, all the entries of liabilities and resources, and the principal correspondents, as well as some other details. The liabilities section includes paid-up capital, surplus and profits, deposits, and "other liabilities." The resources section includes loans and discounts, bonds and securities, cash, and "miscellaneous."⁶ My collected data extends a few periods between 1930 and 1940. The locations and number of banks per location can be seen in can be seen in Figure 3.

2.2 Individual and Household-level data

2.2.1 Data used for place effects

I use the full Count Decennial Census data from IPUMS for 1930 and 1940 which contain both individual level and household level outcomes. As individual level outcomes, I use mainly occupation, migration, and demographic related outcomes. With respect to labor outcomes, I have, for example, employment status, labor force status, weeks work last year, hours work last year, among others. Importantly I also have income information for at least one of the censuses. The census also

⁵Some recent examples are [Das et al. \(2022\)](#), [Jaremski and Wheelock \(2017\)](#), [Mitchener and Richardson \(2019\)](#), [Rajan and Ramcharan \(2015\)](#), [Carlson and Mitchener \(2009\)](#), [Carlson et al. \(2022\)](#), and [Richardson and Troost \(2009\)](#).

⁶See Table 22 for a typical bank example in the data that I collected.

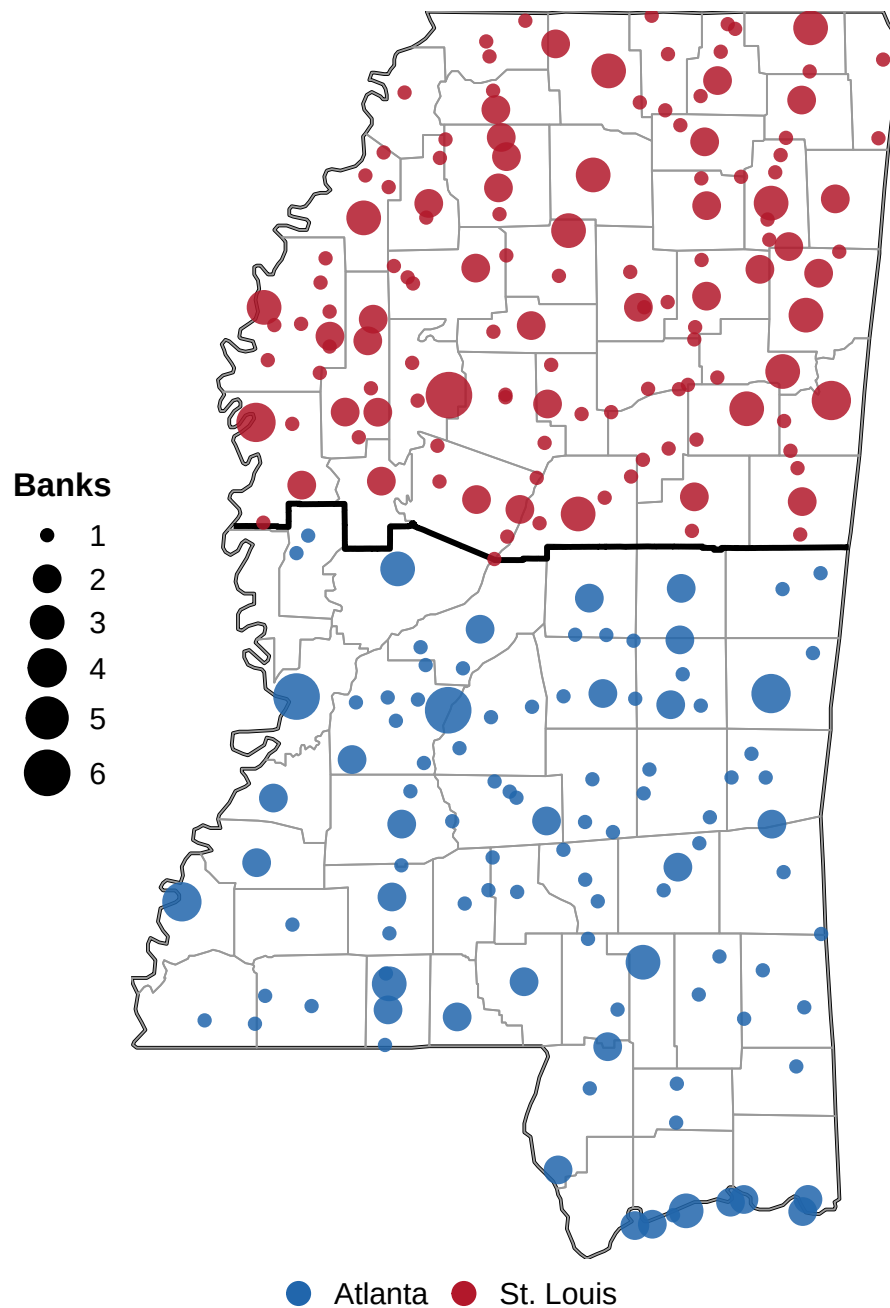


Figure 3. All Mississippi Banks in 1930. The thick line is the boundary between Federal Reserve Districts. Source: Rand McNally's Banker's Directory.

has information about demographic details like, for example, sex, age, marital status, race, Hispanic origin, among others. Finally, among the census' household level outcomes I have information mostly on housing, for example, ownership status, monthly rent, and house value.

An important advantage in my setting is that I have the locations for most of the individuals in Mississippi. In particular, I use the Census Place Project (CPP) [Berkes et al. \(2023\)](#) which provide three important things, one, the location of individuals, and two, a consistent location over censuses, and three, the location is sub-county level.⁷ Specifically, for Mississippi, I have coverage of about 75% of individuals. I show in the Appendix (Figures 16 and 17) that the geo-referenced individuals in Mississippi have essentially the same spatial distribution than the ones that don't have a sub-county location.

2.2.2 Data used for people effects

To study people effects, migration, as well as to compare place v. person effects, I build a panel with the 1930 and 1940 full count censuses by leveraging [Abramitzky et al. \(2021\)](#) crosswalks. In [Abramitzky et al. \(2021\)](#) paper they provide different versions of the crosswalks that use different algorithms. In particular, I use the cross walks that they call "exact standard" because, in their paper, they show that for inference purposes, this is very close to the "ground truth" estimates of inter-generational mobility. As I study other labor market outcomes, it seems fitting based on their results.

In a few words, the algorithm matches only people than can be uniquely identified by name, place of birth, and birth year. If, for example, there are two Jon Snow's that come from the same place of birth, they discard both. So, it is exact in the sense that they have to be uniquely identifiable in those variables. However—

⁷[Berkes et al. \(2023\)](#) have a coverage of more than 90% of individuals (on average) starting from the Census of 1980 and afterwards.

and importantly—the algorithm is otherwise fully automated. So, there is no humane intervention further than the initial conditions set on the algorithm.

In practice, I use simultaneously both the [Abramitzky et al. \(2021\)](#) and the [Berkes et al. \(2023\)](#) crosswalks to construct the panel. After this double-matching, I get a sample that amounts to about 14% of the total USA observations, and around 10% of Mississippi’s observations. Also, this sample contains essentially only males, due to limitations of [Abramitzky et al. \(2021\)](#) algorithm.

3 Methods

As the policy exposure of a bank changes discontinuously at the boundary between the Federal Reserve districts of Atlanta and St. Louis in Mississippi, a method that arises naturally is the Regression Discontinuity Design (RDD). These type of designs had become increasingly popular in recent years especially in the applied microeconomics literature,⁸ one recent example in the macroeconomics-oriented literature is [Carlson et al. \(2022\)](#). Specifically here I use RDD with randomization inference near the districts boundary. Now I explain the method in some detail following [Cattaneo et al. \(2015\)](#) and [Cattaneo et al. \(2016\)](#).

3.1 Data-driven border delimitation

A crucial part of randomization inference in RD designs is the selection of the optimal window around the threshold (see [Cattaneo et al. \(2016\)](#), and [Cattaneo et al. \(2015\)](#)). I follow the data-driven procedure proposed by [Cattaneo et al. \(2015\)](#), which essentially tests the assumption that within such window, the data were generated in an as-if-random manner. Mechanically speaking, the idea is to exploit either the ex-ante or the ex-post unaffected covariates, starting from the

⁸Some reviews are covered in [Lee and Lemieux \(2010\)](#), and more recently in [Cattaneo and Titiunik \(2022\)](#).

smallest possible window that satisfies a minimum number of observations on each side of the threshold. The window is then gradually increased until the null hypothesis is rejected with high probability. Cattaneo et al. (2024) recommend to be conservative and use $\alpha = 0.15$, which I thus use. The selected data-driven optimal boundary is thus the biggest window such that the p -value of the test of balanced covariates is still above 0.15.

3.2 Randomization Inference, a brief overview

Randomization inference is based on the idea that the source of randomness comes only from the treatment assignment itself.⁹ To be more precise, say that the treatment assignment is random in the window $\mathcal{W}$. Let's denote $\mathbf{T}_{\mathcal{W}}$ the random vector of treatment assignment for all units in the window $N_{\mathcal{W}}$. Let also $\mathbf{t}$ denote a specific realization of all the possible treatment vectors $\mathcal{T}_{\mathcal{W}}$. Furthermore, let $N_{\mathcal{W}+}$ and $N_{\mathcal{W}-}$ denote the treated and non-treated units in $\mathcal{W}$, so that $N_{\mathcal{W},r} + N_{\mathcal{W},l} = N_{\mathcal{W}}$. Then the probability that we observe an arbitrary treatment vector $\mathbf{t}$ is:

$$\mathbb{P}_{\mathcal{W}}[\mathbf{T}_{\mathcal{W}} = \mathbf{t}] = \frac{1}{\binom{N_{\mathcal{W}}}{N_{\mathcal{W},r}}} \quad (3.1)$$

where $\binom{N_{\mathcal{W}}}{N_{\mathcal{W},r}}$ represents combinations. Notice that in the case of binary treatment assignment, letting 1 to represent treated and 0 non-treated, $\mathcal{T}_{\mathcal{W}}$ contains all possible indicator variables with $N_{\mathcal{W},r}$ number ones, and $N_{\mathcal{W},l}$ number zeros. In this context, inside $\mathcal{W}$, the number of treated and non-treated units is kept fixed. This is called, a fixed-margins randomization mechanism. Thus, all randomness comes only from *not* knowing which $\mathbf{t} \in \mathcal{T}_{\mathcal{W}}$ we will observe.

By exploiting the assumption of random treatment assignment in $\mathcal{W}$ we can perform hypotheses tests of the form $H_0^F : Y_i(0) = Y_i(1)$ for all units i for which

⁹This subsection follows Cattaneo et al. (2015) and Cattaneo et al. (2024), see these references for further details.

$R_i \in \mathcal{W}$. Here R_i represents the running variable, and $Y_i(0)$, $Y_i(1)$ the potential outcomes for unit i .

To test the hypothesis the procedure is as follows. First, let's take an statistic S , which is a function of $\mathbf{T}_{\mathcal{W}}$ and $\mathbf{Y}_{\mathcal{W}}$, the latter represents the observed outcomes for all units in $\mathcal{W}$. Now, let $S^{\text{obs}} = S(\mathbf{t}_{\mathcal{W}}^{\text{obs}}, \mathbf{Y}_{\mathcal{W}})$ denote the observed value of said statistic which is a function of the observed treatment vector $\mathbf{t}_{\mathcal{W}}^{\text{obs}}$ and the observed outcomes. The finite-sample exact randomization distribution of S comes from the randomness of the treatment assignment. In particular, because under the null all potential outcomes are known, we can calculate S for every possible permutation of the observed treatment assignment $\mathbf{t}_{\mathcal{W}}^{\text{obs}}$. Then we can compute the finite-sample exact p-value as follows:

$$p^{\text{F}} = \mathbb{P}_{\mathcal{W}} \left(S(\mathbf{T}_{\mathcal{W}}, \mathbf{Y}_{\mathcal{W}}) \geq S^{\text{obs}} \right)$$

Then, to build confidence intervals (CI) for the treatment effect, we have to take a grid of nulls and repeat the process for all of them. Collecting all the p-values in this way allows us to invert the tests to get the desired confidence intervals.¹⁰

The randomization approach to inference works for any test statistic, but the one used here is not the conventional, difference in means statistic. I'll describe an extension I propose in the next subsection that deals with clustering in the error terms.

3.3 Randomization Inference under Clustering

Because individuals are agglomerated in towns, or more generally places, a worry might be that the DGP that generates the individual level data might display some form of clustered dependence at the place level. I thus, extend the baseline Randomization Inference to the case of clustered errors. The essence of inference

¹⁰For more details on this test inversion procedure see [Cattaneo et al. \(2024\)](#)

stays the same, as the randomization (permutation) is only of over the treatment assignment itself (ti.e., the elements of the observed treatment vector). However, in contrast to conventional randomization inference in RD designs, and based on Monte Carlo simulations, instead of the baseline statistic, I have to use a studentized-t statistic that at each permutation uses the cluster-robust covariance matrix in its computation. By doing so, I achieve correct size (see Figure 18).

In particular, each permutation the estimated errors are cluster robust following [Cameron et al. \(2011\)](#). To be precise, I use the following specification. For unit $i = 1, \dots, N$, with treatment indicator $D_i \in \{0, 1\}$ and segment $s(i)$,

$$Y_i = \alpha + \tau D_i + \sum_{s=1}^{S-1} \eta_s \mathbf{1}\{s(i) = s\} + \varepsilon_i,$$

where the segment fixed effects are included as a full set of segment indicators (one omitted). For testing the null of no treatment effect, I use the following test statistic,

$$t_{\text{obs}}^{\text{ind}}(\tau_0) = \frac{\hat{\tau} - \tau_0}{\sqrt{e_{\tau}^{\top} \hat{V}_{\text{CL}}(\hat{\beta}) e_{\tau}}}$$

with e_{τ} for the selector of τ , and $\hat{V}_{\text{CL}}$ is the matrix of cluster robust standard errors. Typically, HC1 or HC2. Besides the statistic, the reminder of the randomization inference approach stays the same.

3.3.1 Randomization Inference to compare Place v. People effects

Finally, to test whether the differential policies affected places and people differently, I will use the following specification. I begin by stacking two samples: the places sample (all individuals residing in Mississippi in 1940, abbreviated as “MS-1940”), which is indicated with $G_i = 0$, and the people sample (matched individuals observed in both the 1930 and 1940 Censuses who were residing in Mississippi in

1930, regardless of their 1940 location, and abbreviated here as “matched”), this sample will be indicated with $G_i = 1$. Then I estimate the following equation:

$$Y_i = \alpha + \beta D_i + \theta G_i + \delta (D_i \times G_i) + \sum_{s=1}^{S-1} \eta_s \mathbf{1}\{s(i) = s\} + \varepsilon_i$$

here β represents the *place effect*, $\beta + \delta$ the *people effect*, and δ represents the difference between them. This difference, δ , is the key parameter of interest.

The test statistic to be used is given by:

$$t_{\text{obs}}^{\text{cmp}} = \frac{\hat{\delta} - 0}{\sqrt{e_{\delta}^{\top} \hat{V}_{\text{CL}}(\hat{\beta}) e_{\delta}}}$$

where e_{δ} selects the relevant coefficient, and $\hat{V}_{\text{CL}}$ is computed in a cluster-robust fashion (same clustering level and HCK as in the place estimations).

4 Results

4.1 Window selection and validation

As mentioned above, the data-driven window selection is done by exploiting relevant covariates. The covariates I use are the logarithms of all the bank level outcomes (described in the data section) that I collected from the Rand McNally directories. I then follow [Cattaneo et al. \(2015\)](#) and [Cattaneo et al. \(2024\)](#) who recommend to be such that we have at least 10 observations at each side of the cutoff. After executing the algorithm, the resulting optimal window is 67,020

meters (or about 67km, or about 41.87mi).¹¹ The selection procedure, and the selected window, is illustrated in Figure (7).

To validate the window I perform two exercises. First I show that the pre-treatment bank covariates all have null differences (see Table (9)). Second, as an out of sample exercise, I check whether within the optimal window there are any relevant economic differences between the two sides of the boundary. I don't find any significant difference in any labor outcomes. This is also true even if I restrict the sample to only males or only females. I also check household level outcomes and I find no significant differences except for the two rent-related outcomes, one of which only shows a very weak difference. All things considered, I don't see any relevant economic differences that might be related to long-term economic outcomes. All of this can be seen in Tables (10) to (17).

Having validated the window with banking outcomes, labor market outcomes, and household level outcomes, I proceed to the results. All of the results will use the optimal window here obtained.

4.2 Short term effects on Banks

My first result is that the probability that a bank failed *permanently*, during the period of the Caldwell-related crisis, was 14 percentage points lower banks located in the Atlanta district relative to banks in the St. Louis district. This is reported in Table (1). The result is strongly significant regardless of whether the p-value is asymptotic or comes from cluster-robust randomization inference. Importantly, this result is also economically strongly significant as the mean probability of failure is 10 percent. It is worth to highlight that these are permanent bank failures, which are the relevant failures if they were to have any influence in the long-term.

¹¹For reference, in their seminal paper [Richardson and Troost \(2009\)](#) used a bandwidth of a 3 to 4 counties, or approximately between 63 to 85 miles (100 to 135km). The optimal data-driven bandwidth is roughly half the size of the one there used. A possible reason for the smaller window I get, is because of the presence of Jackson at about 50 miles from the border. It is likely difficult to find a town that could balance Jackson on the other side of the boundary.

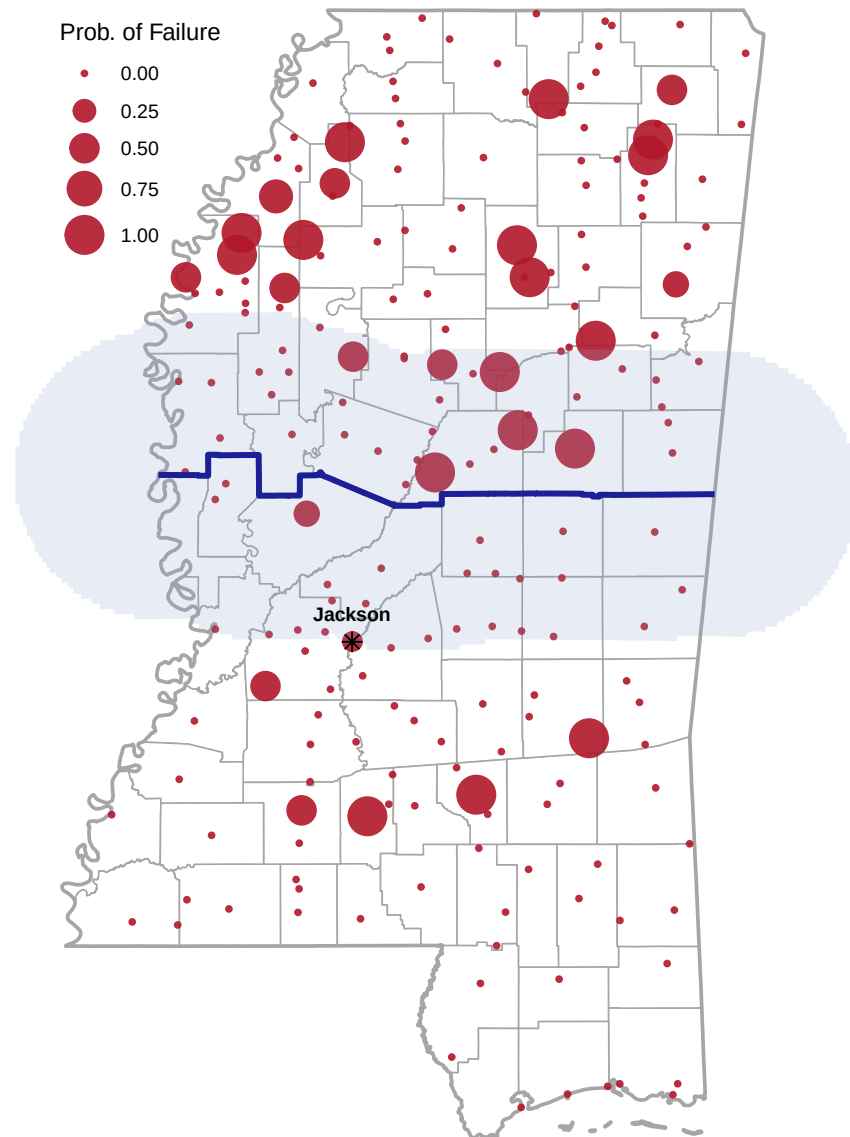


Figure 4. Permanent Bank Failure Probability and Data-driven Optimal Window. The optimal window is 67,020 meters ($\sim 67\text{km}$, $\sim 41.87\text{mi}$).

Indeed I show in Tables 18 and 19 that neither temporary bank failures nor failures in any other banking panic were significantly different between the two districts. A map with the permanent bank failure probability and the optimal window can be seen in Figure 4.

Table 1. 1930: Probability of Permanent Bank Failure

	fail_cald_perm
	(1)
<i>treated</i>	-0.14** (0.07)
P-value	0.03
p (RI)	0.02
Mean (W)	0.10
N	288
Eff. N (T – C)	34 – 55
Clusters	205
Eff. Clusters (T – C)	24 – 36
Segment FE	✓

Notes: The dependent variables are: column (1): Permanently Failed. Window used is 67.02km (41.64mi) from the boundary. Treatment is being on the Atlanta side of the boundary. Errors clustered at the place level where relevant. Segment fixed effects included in the FE specs. Randomization inference (RI) spec used in all columns: RI (individual-level) + FE + Cluster SE. The statistic used is the cluster-robust studentized t with $B = 200$ permutations done at the individual level; the reported t is the observed one; P-value is the asymptotic p-value, while $p(RI)$ is the randomization p-value. Source: own elaboration.

4.3 Main results: long-term effects on Labor Markets

In this subsection I will discuss what were the long-term effects of the Federal Reserve's monetary interventions of late 1930 on labor markets from two perspectives. First, from the local-labor markets perspective which I call place effects because they refer to individuals who live physically in Mississippi in 1940 regardless

of where they were in 1930. Second, from the workers perspective, which I call people, because they refer to the effects on individuals who originally were at the boundary in 1930 and who, ten years later, might be located anywhere in the country.

It's worth mentioning here that the coming results are essentially all unconditional in the sense that they are not restricted to any variable (in particular, my results will not depend on people being part of the labor force, which is in itself endogenously affected by the policy).

4.3.1 Place effects: Effects on Mississippi's Local Labor Market

Labor force participation. I find that, in 1940, for the people living and working in local labor markets in Mississippi, the probability of being in the labor force was 3 percentage points lower in the Atlanta side of the boundary relative to the St Louis side (Table 2). This number is strongly significant, even under randomization inference. The number represents about 5% of the mean, which is economically relevant as more than half of the people is in the labor force.

Unemployment. There is also a statistically significant effect on the unemployment probability. The probability of being unemployed was 1 percentage point lower in Atlanta local labor markets relative to St Louis', or -2 if we include Public Emergency Work as part of the definition of unemployment, as the *official* census does. Regardless of the unemployment definition, the estimate represents about half of the mean for this outcome, and thus it is also very significant economically speaking (see Table 2 and Figure (5)).

It's worth mentioning here that the two results can be reconciled by the fact that a contractionary monetary policy shock can lead to a significant increase in the labor supply [Graves et al. \(2023\)](#). Thus, from the St. Louis district perspective, the contractionary shock meant higher and persistent unemployment, and thus this motivated people to increase job-seeking behavior. This also aligns

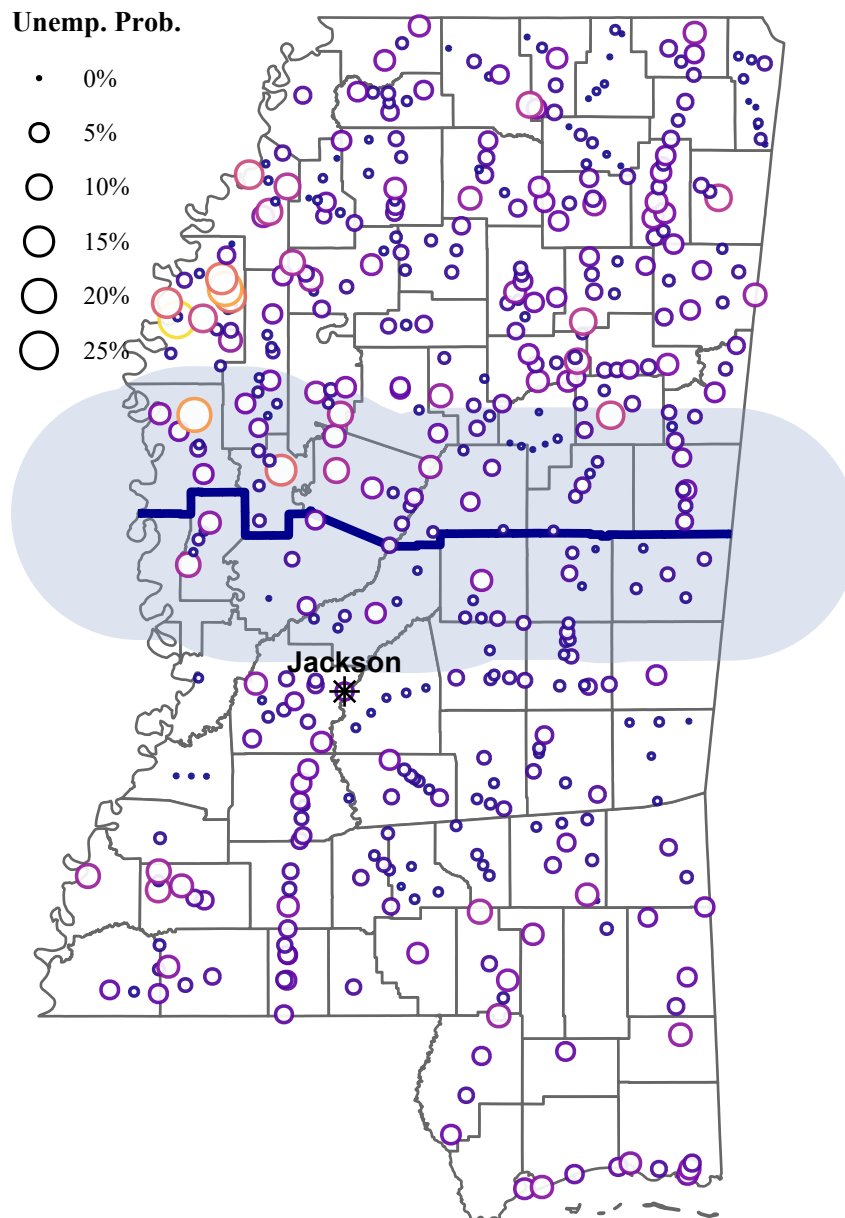


Figure 5. Probability of being unemployed in 1941. In this figure, the definition of Unemployment excludes PEW.

Table 2. 1940: Labor Force Participation and Unemployment Status

	INLABFORCE	UNEMP1	UNEMP2
	(1)	(2)	(3)
<i>treated</i>	-0.03** (0.01)	-0.01* (0.01)	-0.02** (0.01)
P-value	0.02	0.06	0.02
p (RI)	0.04	0.06	0.02
Mean (W)	0.54	0.02	0.04
N	1,108,058	1,108,058	1,108,058
Eff. N (T – C)	177,689 – 210,386	177,689 – 210,386	177,689 – 210,386
Clusters	438	438	438
Eff. Clusters (T – C)	51 – 75	51 – 75	51 – 75
Segment FE	✓	✓	✓

Notes: The dependent variables are: column (1): In Labor Force, column (2): Unemployed (Doesn't include PEW), column (3): Unemployed (Includes PEW). Window used is 67.02km (41.64mi) from the boundary. Treatment is being on the Atlanta side of the boundary. Errors clustered at the place level where relevant. Segment fixed effects included in the FE specs. Randomization inference (RI) spec used in all columns: RI (individual-level) + FE + Cluster SE. The statistic used is the cluster-robust studentized t with $B = 200$ permutations done at the individual level; the reported t is the observed one; P-value is the asymptotic p-value, while p (RI) is the randomization p-value. Source: own elaboration. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

with the ample literature on the added worker effect, which states that when the head of the family loses its job, this motivates other people in her household to start the job search, thus adding himself into the labor force.

Employment. I don't find a significant difference in the probability of being employed in Atlanta relative to St Louis. Indeed, this result is strongly non-significant regardless of whether PEW is included or not as part of the definition of employed. This can be seen in Table 3.

Hours worked. Similarly, in the same Table 3 I report that I don't find any significant differential effects on the hours employed people worked in the reference week (March of 1940). This result is strongly non-significant (p-val. of 0.8).

Income Wages. In Table 4 I report the estimated response of income wages in Mississippi's local labor markets. Due to the fact that for the 1940 Census the

Table 3. 1940: Hours Worked and Employment Status

	HRSWORK1	EMP1	EMP2
	(1)	(2)	(3)
<i>treated</i>	-0.15 (0.74)	-0.02 (0.01)	-0.01 (0.01)
P-value	0.84	0.17	0.55
p (RI)	0.81	0.21	0.57
Mean (W)	20.05	0.52	0.50
N	1,108,058	1,108,058	1,108,058
Eff. N (T – C)	177,689 – 210,386	177,689 – 210,386	177,689 – 210,386
Clusters	438	438	438
Eff. Clusters (T – C)	51 – 75	51 – 75	51 – 75
Segment FE	✓	✓	✓

Notes: The dependent variables are: column (1): Hours Worked, column (2): Employed (Includes PEW), column (3): Employed (Doesn't PEW). Window used is 67.02km (41.64mi) from the boundary. Treatment is being on the Atlanta side of the boundary. Errors clustered at the place level where relevant. Segment fixed effects included in the FE specs. Randomization inference (RI) spec used in all columns: RI (individual-level) + FE + Cluster SE. The statistic used is the cluster-robust studentized t with $B = 200$ permutations done at the individual level; the reported t is the observed one; P-value is the asymptotic p-value, while p (RI) is the randomization p-value. Source: own elaboration. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

variable that measures income wages in 1939 included both income generated through private employers or through PEW, I report income wages for the total sample and for those in the sample that are reporting PEW income. Overall, I don't find significant differences in annual income when all the sample is used. However, if we focus on the income wages for those individuals I identify as very likely in PEW (in 1939), there is a strong and negative effect on annual wages of 2.8 dollars in Atlanta relative to St Louis PEW workers. The number represents almost half of the mean, and thus it is economically significant.¹²

Weeks Worked in 1939. I find a highly significant negative differential effect on weeks worked the year prior to the Census, for individuals in Atlanta relative to the St Louis district, both overall and also if we only focus the PEW population. For the overall sample, the estimate of -2 weeks represents about 8.3% of the mean,

¹²It is important to note that the means are small because the results are unconditional and thus there are many zeroes.

Table 4. 1940: Income Wages in 1939, Overall and PEW

	INCWAGE	WKSWORK1	INCWAGE.PEW1939	WEEKSPEW1939
	(1)	(2)	(3)	(4)
<i>treated</i>	25.32 (30.51)	-2.02*** (0.68)	-2.77** (1.34)	-0.31** (0.13)
P-value	0.41	0.00	0.04	0.02
p (RI)	0.43	0.00	0.03	0.00
Mean (W)	191.47	23.99	5.66	0.52
N	1,108,058	1,108,058	1,108,058	1,108,058
Eff. N (T – C)	177,689 – 210,386	177,689 – 210,386	177,689 – 210,386	177,689 – 210,386
Clusters	438	438	438	438
Eff. Clusters (T – C)	51 – 75	51 – 75	51 – 75	51 – 75
Segment FE	✓	✓	✓	✓

Notes: The dependent variables are: column (1): Wage Income in 1939 (includes PEW), column (2): Weeks Worked in 1939 (includes PEW), column (3): Wage Income in 1939 (only PEW), column (4): Weeks Worked in 1939 (only PEW). Window used is 67.02km (41.64mi) from the boundary. Treatment is being on the Atlanta side of the boundary. Errors clustered at the place level where relevant. Segment fixed effects included in the FE specs. Randomization inference (RI) spec used in all columns: RI (individual-level) + FE + Cluster SE. The statistic used is the cluster-robust studentized t with $B = 200$ permutations done at the individual level; the reported t is the observed one; P-value is the asymptotic p-value, while p (RI) is the randomization p-value. Source: own elaboration. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

and for the PEW sample it represents around 60% of the mean. Notice that the mean for weeks worked is half a week, which is the average over all the sample not only the PEW individuals.

These results must be taken jointly with the result on weeks worked for 1939 (columns (1) and (3) of Table 4). Both are about the exact same period and correspond exactly to the specific sample (all or PEW). For the overall sample, the results indicate that workers in the Atlanta side of the boundary were working less weeks but still received the same overall yearly income. This suggests that the income per week was higher in the Atlanta district relative to the St Louis district.

The results for the PEW sample suggest that the Atlanta district workers needed to work less in PEW than their St Louis counterparts. The negative estimate on PEW makes sense since this is overall PEW income over the year. As the Atlanta workers needed to work less in PEW, the total income generated directly by PEW was also smaller. Since both results (weeks and income) in the PEW sample represent about half of the mean, this suggests that the weekly income wage by working in PEW was likely similar to the St Louis side.

Detailed Employment Status Results

I also include the detailed effects on all the components of the Employment Status (see Table 20). The two more notable results here are columns 2 and 5. In column 2 we can see that there is a very significant effect on being working on Public Emergency jobs, whereas people on the Atlanta side of the boundary were 1 percentage point less likely to work on PEW jobs, that number represents half of the mean. This is consistent with the result on less weeks worked in PEW. Thus, 1940 Mississippi's residents were less likely to have PEW jobs and, at the same time, they worked less hours as PEW. This is likely a reflection of the fact that the St. Louis district was more negatively affected economically than the Atlanta district, and thus, the Government likely implemented PEW programs more frequently on those affected areas. I show the probability of being in PEW in Figure 6. Indeed, the figure suggests that the areas on the St. Louis district area had workers more likely to be on PEW.

In addition, in column 5, we can see that the result on unemployment is driven mainly by unemployed workers with experience in the labor market, which matters because this means the results are not driven by new workers entering the labor force.

4.3.2 People Effects: Effects on Workers

Labor Force Participation and Employment. I show in the panel A of Tables (5 and 6) what seems to be a common factor among labor market outcomes for workers who were originally at the boundary between districts. As can be seen, there are not discernable differences in labor force participation, nor in employment probability, regardless of whether we include PEW in the definition of employment. The latter result was expected due to the strong null effects on places' employment, but the null result on labor force participation is really strong (p-value of 0.95 or more).

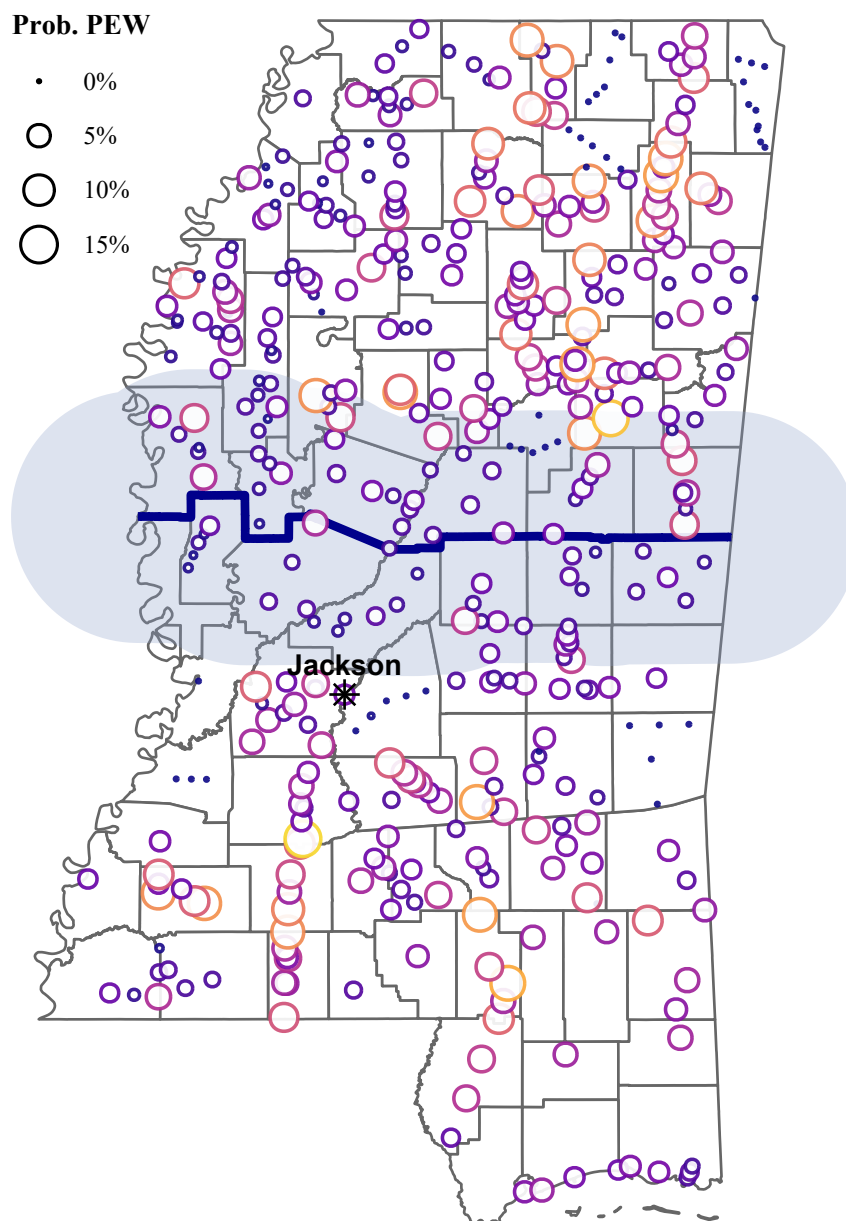


Figure 6. Probability of being in Work Relief

Table 5. Place vs people: Unemployment and Participation

	INLABORCE	UNEMP1	UNEMP2
	(1)	(2)	(3)
Panel A: People effects (Matched)			
<i>treated</i>	-0.00 (0.01)	-0.01 (0.00)	-0.01** (0.00)
P-value	0.95	0.11	0.03
p (RI)	0.97	0.14	0.05
Mean (W)	0.80	0.04	0.07
N	125,800	125,800	125,800
Eff. N (T – C)	21,602 – 20,717	21,602 – 20,717	21,602 – 20,717
Clusters	463	463	463
Eff. Clusters (T – C)	53 – 80	53 – 80	53 – 80
Segment FE	✓	✓	✓
Panel B: Place vs People (D×G)			
<i>treated</i> × G	0.03** (0.02)	0.01* (0.00)	0.01* (0.01)
P (1-sided)	0.03	0.08	0.09
p (RI, 1-sided)	0.04	0.10	0.12
Eff. N (T – C)	199,291 – 231,103	199,291 – 231,103	199,291 – 231,103
Eff. Clusters (T – C)	104 – 155	104 – 155	104 – 155
Segment FE	✓	✓	✓

Notes: Panel A (People effects): estimates from the 1930–1940 matched sample (covariates/treatment/clustering at 1930, outcomes at 1940). Panel B (Place vs People): coefficient on *treated* × G comparing MS 1940 (G=0) vs matched (G=1). Dependent variables are column (1): In Labor Force, column (2): Unemployed (No PEW), column (3): Unemployed (Includes PEW). Treatment is being on the Atlanta side of the boundary. Window is 67.02km (41.64mi) from the boundary. Errors clustered at the place level level; segment fixed effects included where indicated. Randomization inference uses a cluster-robust studentized *t* with *B* = 200 permutations. Panel B reports one-sided p-values (asymptotic and RI). People-effects regressions use 1940 outcomes with 1930 treatment, running variable, clusters, and fixed effects. Place-vs-People regressions use that structure for the matched sample, while the MS-1940 sample uses 1940 variables for all components. Statistical significance: *** *p* < 0.01, ** *p* < 0.05, * *p* < 0.10. Source: own elaboration.

Unemployment. Still, there are strong significant differences in unemployment probability if we include PEW in its definition. This number, even though similar in magnitude to the one observed in places, its much smaller relative to the mean (about 14% instead of around 50% in the case of places). In principle, this seemingly is the only difference that persists for people. Meaning, the people originally at the boundary who were in the Atlanta side, were 1 percentage point less likely to be unemployed in 1940 regardless of where in the country they were. Thus, regardless of whether we care about unemployment at the place level, or at the individual level, monetary interventions affected these outcomes for at least 10 years. This is a novel result. Most evidence about the effect of monetary policy on

unemployment points to effects that lasts for up to 5 years (e.g. [Romer and Romer \(2023\)](#)). The only other paper that discusses long term effects on labor outcomes is [Jordà et al. \(2024\)](#) and does so only for aggregate, international data.

4.3.3 Place vs Person Effects

Table 6. Place vs people: Employment

	EMP1	EMP2
	(1)	(2)
Panel A: People effects (Matched)		
<i>treated</i>	0.00 (0.01)	0.01 (0.01)
P-value	0.42	0.12
p (RI)	0.40	0.16
Mean (W)	0.77	0.73
N	125,800	125,800
Eff. N (T – C)	21,602 – 20,717	21,602 – 20,717
Clusters	463	463
Eff. Clusters (T – C)	53 – 80	53 – 80
Segment FE	✓	✓
Panel B: Place vs People (D × G)		
<i>treated</i> × G	0.02* (0.02)	0.02* (0.02)
P (1-sided)	0.07	0.10
p (RI, 1-sided)	0.07	0.15
Eff. N (T – C)	199,291 – 231,103	199,291 – 231,103
Eff. Clusters (T – C)	104 – 155	104 – 155
Segment FE	✓	✓

Notes: Panel A (People effects): estimates from the 1930–1940 matched sample (covariates/treatment/clustering at 1930, outcomes at 1940). Panel B (Place vs People): coefficient on *treated* × G comparing MS 1940 (G=0) vs matched (G=1). Dependent variables are column (1): Employed (Includes PEW), column (2): Employed (Doesn't include PEW). Treatment is being on the Atlanta side of the boundary. Window is 67.02km (41.64mi) from the boundary. Errors clustered at the place level level; segment fixed effects included where indicated. Randomization inference uses a cluster-robust studentized *t* with *B* = 200 permutations. Panel B reports one-sided p-values (asymptotic and RI). People-effects regressions use 1940 outcomes with 1930 treatment, running variable, clusters, and fixed effects. Place-vs-People regressions use that structure for the matched sample, while the MS-1940 sample uses 1940 variables for all components. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Source: own elaboration.

As we saw, the effects on people are much more muted than the effects on places. Are these differences significant? I show in panel B of Tables (5 and 6) the results of testing precisely that null. As can be seen, in all cases the differences are significant at the very least using a one-sided test (although in some cases regardless of whether a two-sided test is used). Furthermore, these results show that in every case the people effects go in the opposite direction than the place effects. People were able to counteract most of the place effects. The only surviving difference is, as mentioned above, the unemployment probability result. But even in this case, people originally in the St. Louis side of the boundary were able to close the gap considerably relative to people originally on the Atlanta side.

4.3.4 Migration

The above results suggest that people counteracted the relatively contractionary policy of the St. Louis district. One natural channel to do so would be migration. Indeed, I show in Table 7 that individuals that were in the Atlanta side of the boundary were 2 percentage points less likely to migrate out of Mississippi and 5 percentage points less likely to migrate out of its county of origin. These results are strongly significant and represent 11% of the mean, and 10.4% of the mean respectively. Thus, both of these results show consistently that the residents of the St Louis side of the boundary migrated more.

5 Closing Remarks

In this paper I studied the short-term effects on banks and the long-term effects on labor markets resulting from the heterogeneous Federal Reserve policy responses to the 1930–31 bank failures in Mississippi. I exploit one of the clearest natural experiments in the history of (unconventional) monetary policy, i.e. when the St. Louis and Atlanta Fed districts adopted sharply different responses to the

Table 7. 1930 to 1940 Out-migration

	outstatemigrant	outcountymigrant
	(1)	(2)
<i>treated</i>	-0.02** (0.01)	-0.05*** (0.02)
P-value	0.03	0.01
p (RI)	0.06	0.01
Mean (W)	0.18	0.48
N	125,800	125,800
Eff. N (T – C)	21,602 – 20,717	21,602 – 20,717
Clusters	463	463
Eff. Clusters (T – C)	53 – 80	53 – 80
Segment FE	✓	✓

Notes: People-effects regressions use 1940 outcomes with 1930 treatment, running variable, clusters, and fixed effects. Dependent variables are column (1): Out-of-State Migrant, column (2): Out-of-County Migrant. Treatment is being on the Atlanta side of the boundary. Window is 67.02km (41.64mi) from the boundary. Errors clustered at the place level level; segment fixed effects included where indicated. Randomization inference uses a cluster-robust studentized t with $B = 200$ permutations. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Source: own elaboration.

Caldwell & Co.-related shock, due largely to differences in beliefs. I show that the more expansionary stance of the Atlanta Fed significantly reduced the probability of permanent bank failures in the short run, while in the long run, this policy divergence had persistent effects on labor markets, both from place and people's perspective.

On local labor markets, Mississippi residents ten years later were less likely to be unemployed, less likely to be in work relief, and earned better weekly wages. On people the effects were much more muted. Only the response on unemployment rate survived, which strongly suggests people were able to counteract the negative shock that persisted on St. Louis places. I reconcile this by showing that people originally in the St. Louis side of the boundary were more likely to migrate.

For today's policymakers, the main lesson is clear: in the face of sudden disruptions to the banking system, monetary contraction can generate long-term damage on local labor markets and workers, unless offset by active monetary interventions.

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6 Appendix A: Main Result Tables

Table 8. Characteristics of Banks in Mississippi, 1930

Variable	Mississippi			Atlanta District			St. Louis District		
	Mean	SD	Median	Mean	SD	Median	Mean	SD	Median
Paid-up capital	53.9	69.6	30.0	62.2	83.4	30.0	47.5	56.2	26.2
Surplus and profits	49.4	93.8	20.0	63.8	118.1	24.4	38.4	68.1	19.1
Deposits	823.9	1182.3	468.9	1044.6	1432.5	531.4	654.7	916.2	382.9
Other liabilities	42.8	102.0	4.1	57.5	131.3	9.0	31.5	70.3	2.8
Loans and discounts	545.9	846.8	290.0	690.6	1007.3	334.1	434.9	682.0	253.9
Bonds and securities	184.0	286.3	84.7	256.8	375.2	110.7	128.6	175.1	54.5
Miscellaneous assets	43.3	81.1	14.6	51.9	99.6	14.9	36.7	63.0	14.0
Cash and exchange	196.4	286.0	101.6	229.4	333.9	96.7	171.1	240.9	107.9
Total assets	970.2	1394.2	528.6	1231.6	1719.6	593.5	771.3	1046.2	489.5

Note: All values are in thousands of dollars. Summary statistics computed for all banks in operation as of November 1930. Source: Rand-McNally Bankers Directory and author's compilation.

6.1 Window validation: covariates

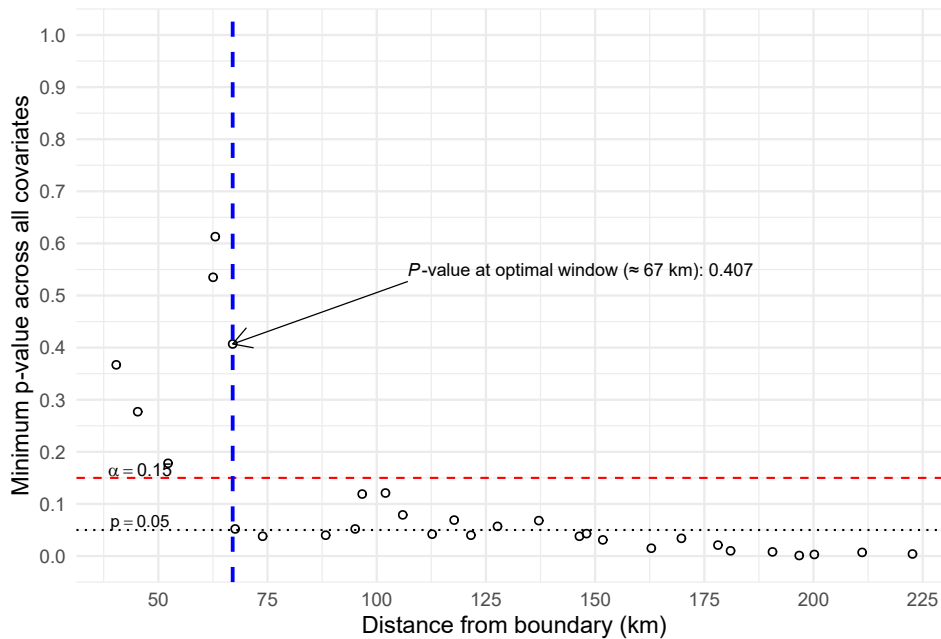


Figure 7. Window selection based on predetermined covariates. In each window, for each covariate, a randomization-based test is performed for the sharp null hypothesis of no treatment effect. The p-values indicate the smallest p-value across all covariates. The significance level used to choose the window was $\alpha = 0.150$; the minimum number of observations per side was 15; the number of repetitions was 1000; I evaluated 30 candidate windows spanning all the support; the window grid used increments of at least 2 observations per side; the window was constrained to be symmetric; the kernel was uniform; the covariate balance test used was difference in means. Distances are in kilometers (miles in parentheses). Optimal half-window ≈ 67 km (41.6 mi).

Table 9. Window validation, pre-treatment bank-level placebos

	log_assets	log_deposits	log_paid_up_capital	log_surplus_profits	log_bonds_securities	log_pop_perbank
	(1)	(2)	(3)	(4)	(5)	(6)
<i>treated</i>	0.19 (0.33)	0.21 (0.34)	0.15 (0.28)	0.22 (0.36)	0.97 (0.89)	0.12 (0.29)
P-value	0.56	0.53	0.59	0.53	0.28	0.69
p (RI)	0.57	0.55	0.60	0.54	0.32	0.73
Mean (W)	13.27	13.10	10.49	10.12	10.15	6.75
N	288	288	288	288	288	288
Eff. N (T – C)	34 – 55	34 – 55	34 – 55	34 – 55	34 – 55	34 – 55
Clusters	205	205	205	205	205	205
Eff. Clusters (T – C)	24 – 36	24 – 36	24 – 36	24 – 36	24 – 36	24 – 36
Segment FE	✓	✓	✓	✓	✓	✓

Notes: The dependent variables are: column (1): Assets, column (2): Deposits, column (3): Paid-up capital, column (4): Surplus and profits, column (5): Bonds and securities, column (6): Population per bank. Window used is 67.02km (41.64mi) from the boundary. Treatment is being on the Atlanta side of the boundary. Errors clustered at the place level where relevant. Segment fixed effects included in the FE specs. Randomization inference (RI) spec used in all columns: RI (individual-level) + FE + Cluster SE. The statistic used is the cluster-robust studentized t with $B = 200$ permutations done at the individual level; the reported t is the observed one; P-value is the asymptotic p-value, while p (RI) is the randomization p-value. Source: own elaboration.

6.2 Window validation: 1930 census outcomes

Table 10. 1930: Employment Outcomes

	INLABFORCE	EMP	UNEMP
	(1)	(2)	(3)
<i>treated</i>	-0.01 (0.01)	-0.01 (0.02)	0.00 (0.00)
P-value	0.60	0.51	0.42
p (RI)	0.62	0.56	0.52
Mean (W)	0.63	0.61	0.02
N	936,460	936,460	936,460
Eff. N (T – C)	159,688 – 169,347	159,688 – 169,347	159,688 – 169,347
Clusters	463	463	463
Eff. Clusters (T – C)	53 – 80	53 – 80	53 – 80
Segment FE	✓	✓	✓

Notes: The dependent variables are: column (1): In Labor Force, column (2): Employed, column (3): Unemployed. Window used is 67.02km (41.64mi) from the boundary. Treatment is being on the Atlanta side of the boundary. Errors clustered at the place level where relevant. Segment fixed effects included in the FE specs. Randomization inference (RI) spec used in all columns: RI (individual-level) + FE + Cluster SE. The statistic used is the cluster-robust studentized t with $B = 200$ permutations done at the individual level; the reported t is the observed one; P-value is the asymptotic p-value, while p (RI) is the randomization p-value. Source: own elaboration.

Table 11. 1930: Employment Outcomes for Males

	INLABFORCE	EMP	UNEMP
	(1)	(2)	(3)
<i>treated</i>	-0.01 (0.01)	-0.01 (0.01)	0.01 (0.01)
P-value	0.46	0.38	0.30
p (RI)	0.46	0.36	0.32
Mean (W)	0.92	0.89	0.03
N	936,460	936,460	936,460
Eff. N (T – C)	78,027 – 83,250	78,027 – 83,250	78,027 – 83,250
Clusters	463	463	463
Eff. Clusters (T – C)	53 – 80	53 – 80	53 – 80
Segment FE	✓	✓	✓

Notes: The dependent variables are: column (1): In Labor Force, column (2): Employed, column (3): Unemployed. Window used is 67.02km (41.64mi) from the boundary. Treatment is being on the Atlanta side of the boundary. Errors clustered at the place level where relevant. Segment fixed effects included in the FE specs. Randomization inference (RI) spec used in all columns: RI (individual-level) + FE + Cluster SE. The statistic used is the cluster-robust studentized t with $B = 200$ permutations done at the individual level; the reported t is the observed one; P-value is the asymptotic p-value, while $p(RI)$ is the randomization p-value. Results restricted to Males. Source: own elaboration.

Table 12. 1930: Employment Outcomes for Females

	INLABFORCE	EMP	UNEMP
	(1)	(2)	(3)
<i>treated</i>	-0.01 (0.03)	-0.01 (0.02)	0.00 (0.00)
P-value	0.72	0.69	0.80
p (RI)	0.68	0.64	0.84
Mean (W)	0.36	0.35	0.01
N	936,460	936,460	936,460
Eff. N (T – C)	81,661 – 86,097	81,661 – 86,097	81,661 – 86,097
Clusters	463	463	463
Eff. Clusters (T – C)	53 – 80	53 – 80	53 – 80
Segment FE	✓	✓	✓

Notes: The dependent variables are: column (1): In Labor Force, column (2): Employed, column (3): Unemployed. Window used is 67.02km (41.64mi) from the boundary. Treatment is being on the Atlanta side of the boundary. Errors clustered at the place level where relevant. Segment fixed effects included in the FE specs. Randomization inference (RI) spec used in all columns: RI (individual-level) + FE + Cluster SE. The statistic used is the cluster-robust studentized t with $B = 200$ permutations done at the individual level; the reported t is the observed one; P-value is the asymptotic p-value, while p (RI) is the randomization p-value. Results restricted to restricted to Females. Source: own elaboration.

Table 13. 1930: Employment Outcomes for Black Individuals

	INLABFORCE	EMP	UNEMP
	(1)	(2)	(3)
<i>treated</i>	0.01 (0.02)	-0.00 (0.02)	0.01 (0.01)
P-value	0.82	0.95	0.42
p (RI)	0.80	0.96	0.42
Mean (W)	0.71	0.69	0.02
N	936,460	936,460	936,460
Eff. N (T – C)	78,503 – 100,936	78,503 – 100,936	78,503 – 100,936
Clusters	463	463	463
Eff. Clusters (T – C)	53 – 80	53 – 80	53 – 80
Segment FE	✓	✓	✓

Notes: The dependent variables are: column (1): In Labor Force, column (2): Employed, column (3): Unemployed. Window used is 67.02km (41.64mi) from the boundary. Treatment is being on the Atlanta side of the boundary. Errors clustered at the place level where relevant. Segment fixed effects included in the FE specs. Randomization inference (RI) spec used in all columns: RI (individual-level) + FE + Cluster SE. The statistic used is the cluster-robust studentized t with $B = 200$ permutations done at the individual level; the reported t is the observed one; P-value is the asymptotic p-value, while p (RI) is the randomization p-value. Results restricted to black individuals. Source: own elaboration.

Table 14. 1930: Employment Outcomes for White Individuals

	INLABFORCE	EMP	UNEMP
	(1)	(2)	(3)
<i>treated</i>	-0.00 (0.01)	-0.00 (0.01)	0.00 (0.00)
P-value	0.77	0.89	0.60
p (RI)	0.80	0.92	0.62
Mean (W)	0.54	0.52	0.02
N	936,460	936,460	936,460
Eff. N (T – C)	80,433 – 68,267	80,433 – 68,267	80,433 – 68,267
Clusters	463	463	463
Eff. Clusters (T – C)	53 – 80	53 – 80	53 – 80
Segment FE	✓	✓	✓

Notes: The dependent variables are: column (1): In Labor Force, column (2): Employed, column (3): Unemployed. Window used is 67.02km (41.64mi) from the boundary. Treatment is being on the Atlanta side of the boundary. Errors clustered at the place level where relevant. Segment fixed effects included in the FE specs. Randomization inference (RI) spec used in all columns: RI (individual-level) + FE + Cluster SE. The statistic used is the cluster-robust studentized t with $B = 200$ permutations done at the individual level; the reported t is the observed one; P-value is the asymptotic p-value, while p (RI) is the randomization p-value. Results restricted to white individuals. Source: own elaboration.

Table 15. 1930: Household-level Outcomes

	ISURBAN	ISFARM	OWNER	RENTER	HASRADIO
	(1)	(2)	(3)	(4)	(5)
<i>treated</i>	0.06 (0.12)	-0.03 (0.10)	0.03 (0.02)	-0.04* (0.03)	0.01 (0.02)
P-value	0.61	0.78	0.16	0.09	0.51
p (RI)	0.68	0.80	0.14	0.07	0.52
Mean (W)	0.28	0.56	0.32	0.67	0.07
N	357,927	357,927	357,927	357,927	357,927
Eff. N (T – C)	61,005 – 65,968	61,005 – 65,968	61,005 – 65,968	61,005 – 65,968	61,005 – 65,968
Clusters	463	463	463	463	463
Eff. Clusters (T – C)	53 – 80	53 – 80	53 – 80	53 – 80	53 – 80
Segment FE	✓	✓	✓	✓	✓

Notes: The dependent variables are: column (1): Urban Status, column (2): Farm Status, column (3): Owner of Dwelling, column (4): Renter of Dwelling, column (5): Has Radio. Window used is 67.02km (41.64mi) from the boundary. Treatment is being on the Atlanta side of the boundary. Errors clustered at the place level where relevant. Segment fixed effects included in the FE specs. Randomization inference (RI) spec used in all columns: RI (individual-level) + FE + Cluster SE. The statistic used is the cluster-robust studentized t with $B = 200$ permutations done at the individual level; the reported t is the observed one; P-value is the asymptotic p-value, while p (RI) is the randomization p-value. Source: own elaboration.

Table 16. 1930: Rents

	RENT30_2
	(1)
<i>treated</i>	-4.19** (1.76)
P-value	0.02
p (RI)	0.01
Mean (W)	20.90
N	357,927
Eff. N (T – C)	17,006 – 16,587
Clusters	463
Eff. Clusters (T – C)	53 – 79
Segment FE	✓

Notes: The dependent variables are: column (1): Rent. Window used is 67.02km (41.64mi) from the boundary. Treatment is being on the Atlanta side of the boundary. Errors clustered at the place level where relevant. Segment fixed effects included in the FE specs. Randomization inference (RI) spec used in all columns: RI (individual-level) + FE + Cluster SE. The statistic used is the cluster-robust studentized t with $B = 200$ permutations done at the individual level; the reported t is the observed one; P-value is the asymptotic p-value, while p (RI) is the randomization p-value. Results restricted to dwelling renters, non-farm. Source: own elaboration.

Table 17. 1930: House Values

	VALUEH.2
	(1)
<i>treated</i>	212.44 (261.94)
P-value	0.42
p (RI)	0.50
Mean (W)	3237.84
N	357,927
Eff. N (T – C)	21,547 – 18,620
Clusters	463
Eff. Clusters (T – C)	53 – 80
Segment FE	✓

Notes: The dependent variables are: column (1): House Value. Window used is 67.02km (41.64mi) from the boundary. Treatment is being on the Atlanta side of the boundary. Errors clustered at the place level where relevant. Segment fixed effects included in the FE specs. Randomization inference (RI) spec used in all columns: RI (individual-level) + FE + Cluster SE. The statistic used is the cluster-robust studentized t with $B = 200$ permutations done at the individual level; the reported t is the observed one; P-value is the asymptotic p-value, while p (RI) is the randomization p-value. Results restricted to dwelling owners. Source: own elaboration.

Table 18. Probability of Failure

	fail_cald_perm	failed_cald	failed	fail_perm	fail_temp
	(1)	(2)	(3)	(4)	(5)
<i>treated</i>	-0.14** (0.07)	-0.10 (0.09)	-0.02 (0.12)	-0.11 (0.09)	0.09 (0.07)
P-value	0.03	0.28	0.88	0.25	0.21
p (RI)	0.02	0.30	0.88	0.30	0.22
Mean (W)	0.10	0.18	0.35	0.24	0.11
N	288	288	288	288	288
Eff. N (T – C)	34 – 55	34 – 55	34 – 55	34 – 55	34 – 55
Clusters	205	205	205	205	205
Eff. Clusters (T – C)	24 – 36	24 – 36	24 – 36	24 – 36	24 – 36
Segment FE	✓	✓	✓	✓	✓

Notes: The dependent variables are: column (1): Permanently Failed in Caldwell Crisis, column (2): Failed in Caldwell Crisis, column (3): Failed in Great Depression, column (4): Permanently Failed in Great Depression, column (5): Temporarily failed in Great Depression. Window used is 67.02km (41.64mi) from the boundary. Treatment is being on the Atlanta side of the boundary. Errors clustered at the place level where relevant. Segment fixed effects included in the FE specs. Randomization inference (RI) spec used in all columns: RI (individual-level) + FE + Cluster SE. The statistic used is the cluster-robust studentized t with $B = 200$ permutations done at the individual level; the reported t is the observed one; P-value is the asymptotic p-value, while p (RI) is the randomization p-value. Source: own elaboration.

Table 19. Probability of Permanent Failure by Crisis Window

	PF_PANIC1	PF_PANIC2	PF_PANIC3	PF_PANIC4	PF_PANIC5	PF_PANIC6	PF_PANIC7
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>treated</i>	-0.14** (0.07)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	-0.02 (0.04)	0.00 (0.00)	-0.05 (0.03)
P-value	0.03				0.60		0.17
p (RI)	0.02				0.62		0.33
Mean (W)	0.10	0.00	0.00	0.00	0.04	0.00	0.02
N	288	288	288	288	288	288	288
Eff. N (T – C)	34 – 55	34 – 55	34 – 55	34 – 55	34 – 55	34 – 55	34 – 55
Clusters	205	205	205	205	205	205	205
Eff. Clusters (T – C)	24 – 36	24 – 36	24 – 36	24 – 36	24 – 36	24 – 36	24 – 36
Segment FE	✓	✓	✓	✓	✓	✓	✓

Notes: The dependent variables are: column (1): Caldwell Crisis, column (2): First Chicago Panic, column (3): Crisis After German Panic, column (4): Crisis After England Departs Gold, column (5): Winter 1932 Crisis, column (6): Second Chicago Panic, column (7): Winter 1933 Crisis. Window used is 67.02km (41.64mi) from the boundary. Treatment is being on the Atlanta side of the boundary. Errors clustered at the place level where relevant. Segment fixed effects included in the FE specs. Randomization inference (RI) spec used in all columns: RI (individual-level) + FE + Cluster SE. The statistic used is the cluster-robust studentized t with $B = 200$ permutations done at the individual level; the reported t is the observed one; P-value is the asymptotic p-value, while p (RI) is the randomization p-value. Source: own elaboration.

Table 20. 1940: Place effects, Detailed Employment Status indicators

	EMPSTATD10	EMPSTATD11	EMPSTATD12	EMPSTATD13	EMPSTATD21	EMPSTATD22	EMPSTATD31	EMPSTATD32	EMPSTATD33	EMPSTATD34
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
<i>treated</i>	-0.00 (0.01)	-0.01*** (0.00)	-0.01** (0.00)	-0.00 (0.00)	-0.01** (0.01)	0.00 (0.00)	-0.02 (0.03)	-0.01 (0.01)	-0.01 (0.01)	0.07* (0.04)
P-value	0.90	0.01	0.03	0.75	0.03	0.11	0.38	0.16	0.41	0.09
p (RI)	0.91	0.01	0.06	0.80	0.01	0.08	0.40	0.16	0.38	0.13
Mean (W)	0.49	0.02	0.01	0.00	0.02	0.00	0.24	0.04	0.08	0.09
N	1,108,058	1,108,058	1,108,058	1,108,058	1,108,058	1,108,058	1,108,058	1,108,058	1,108,058	1,108,058
Eff. N (T – C)	177,689 – 210,386	177,689 – 210,386	177,689 – 210,386	177,689 – 210,386	177,689 – 210,386	177,689 – 210,386	177,689 – 210,386	177,689 – 210,386	177,689 – 210,386	177,689 – 210,386
Clusters	438	438	438	438	438	438	438	438	438	438
Eff. Clusters (T – C)	51 – 75	51 – 75	51 – 75	51 – 75	51 – 75	51 – 75	51 – 75	51 – 75	51 – 75	51 – 75
Segment FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Notes: The dependent variables are: column (1): At work, column (2): At work, public emerg, column (3): Has job, not working, column (4): Armed forces, column (5): Unemp, exper worker, column (6): Unemp, new worker, column (7): NILF, housework, column (8): NILF, unable to work, column (9): NILF, school, column (10): NILF, other. Window used is 67.02km (41.64mi) from the boundary. Treatment is being on the Atlanta side of the boundary. Errors clustered at the place level where relevant. Segment fixed effects included in the FE specs. Randomization inference (RI) spec used in all columns: RI (individual-level) + FE + Cluster SE. The statistic used is the cluster-robust studentized t with $B = 200$ permutations done at the individual level; the reported t is the observed one; P-value is the asymptotic p-value, while p (RI) is the randomization p-value. Source: own elaboration.

Table 21. 1940: Place vs People: Employment Status (Detailed)

	EMPSTATD10	EMPSTATD11	EMPSTATD12	EMPSTATD13	EMPSTATD21	EMPSTATD22	EMPSTATD31	EMPSTATD32	EMPSTATD33	EMPSTATD34
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Panel A: People effects (Matched)										
<i>treated</i>	0.01 (0.01)	-0.01** (0.00)	-0.00 (0.00)	0.00*** (0.00)	-0.01** (0.00)	0.00 (0.00)	-0.00* (0.00)	-0.00 (0.00)	-0.00 (0.01)	0.00 (0.01)
P-value	0.15	0.05	0.36	0.01	0.04	0.13	0.09	0.50	0.91	0.60
p (RI)	0.17	0.04	0.39	0.01	0.06	0.15	0.10	0.52	0.91	0.62
Mean (W)	0.72	0.03	0.01	0.01	0.03	0.00	0.01	0.04	0.10	0.05
N	125,800	125,800	125,800	125,800	125,800	125,800	125,800	125,800	125,800	125,800
Eff. N (T – C)	21,602 – 20,717	21,602 – 20,717	21,602 – 20,717	21,602 – 20,717	21,602 – 20,717	21,602 – 20,717	21,602 – 20,717	21,602 – 20,717	21,602 – 20,717	21,602 – 20,717
Clusters	463	463	463	463	463	463	463	463	463	463
Eff. Clusters (T – C)	53 – 80	53 – 80	53 – 80	53 – 80	53 – 80	53 – 80	53 – 80	53 – 80	53 – 80	53 – 80
Segment FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Panel B: Place vs People (D × G)										
<i>treated</i> × <i>G</i>	0.01 (0.02)	0.00 (0.00)	0.01*** (0.00)	0.00*** (0.00)	0.01* (0.00)	-0.00 (0.00)	0.02 (0.03)	0.01* (0.00)	0.01 (0.01)	-0.06 (0.03)
P (1-sided)	0.21	0.17	0.00	0.01	0.07	0.55	0.27	0.06	0.18	0.96
p (RI, 1-sided)	0.20	0.10	0.01	0.01	0.05	0.58	0.23	0.02	0.14	0.97
Eff. N (T – C)	199,291 – 231,103	199,291 – 231,103	199,291 – 231,103	199,291 – 231,103	199,291 – 231,103	199,291 – 231,103	199,291 – 231,103	199,291 – 231,103	199,291 – 231,103	199,291 – 231,103
Eff. Clusters (T – C)	104 – 155	104 – 155	104 – 155	104 – 155	104 – 155	104 – 155	104 – 155	104 – 155	104 – 155	104 – 155
Segment FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Notes: Panel A (People effects): estimates from the 1930–1940 matched sample (covariates/treatment/clustering at 1930, outcomes at 1940). Panel B (Place vs People): coefficient on *treated* × *G* comparing MS 1940 (*G*=0) vs matched (*G*=1). Dependent variables are column (1): At work, column (2): At work, public emergency, column (3): Has job, not working, column (4): Armed forces, column (5): Unemployed, experienced worker, column (6): Unemployed, new worker, column (7): NILF: housework, column (8): NILF: unable to work, column (9): NILF: school, column (10): NILF: other. Treatment is being on the Atlanta side of the boundary. Window is 67.02km (41.64mi) from the boundary. Errors clustered at the place level level; segment fixed effects included where indicated. Randomization inference uses a cluster-robust studentized *t* with *B* = 200 permutations. Panel B reports one-sided p-values (asymptotic and RI). People-effects regressions use 1940 outcomes with 1930 treatment, running variable, clusters, and fixed effects. Place-vs-People regressions use that structure for the matched sample, while the MS-1940 sample uses 1940 variables for all components. Statistical significance: *** *p* < 0.01, ** *p* < 0.05, * *p* < 0.10. Source: own elaboration.

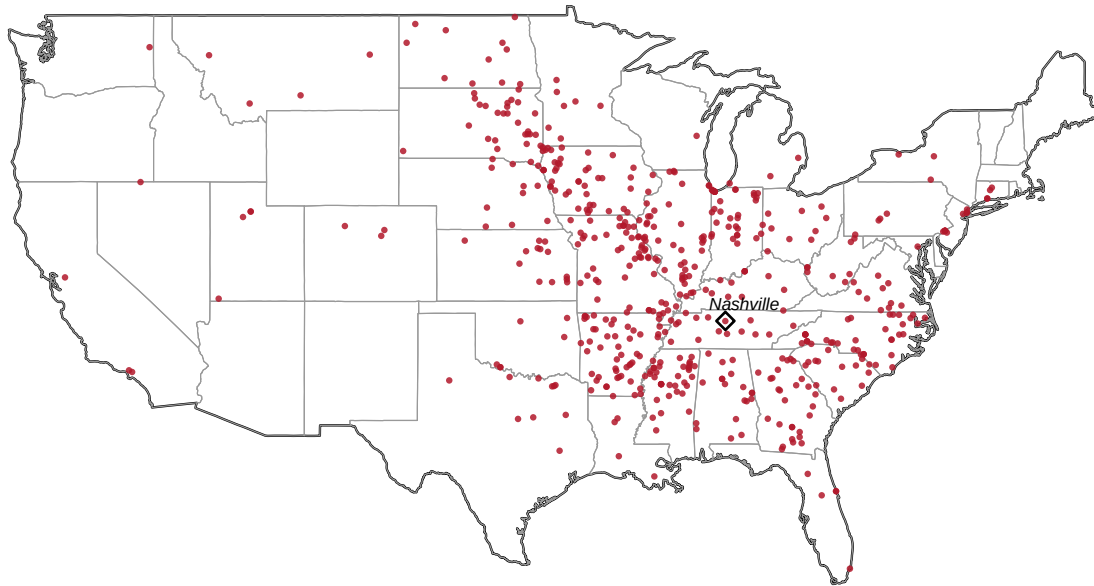


Figure 8. Permanent bank suspensions attributed to the Caldwell collapse. Nashville is where the Bank of Tennessee and Caldwell & Co. were located. The suspensions attributed to the Caldwell collapse follows the banking panics categorization of [Mitchener and Richardson \(2019\)](#).

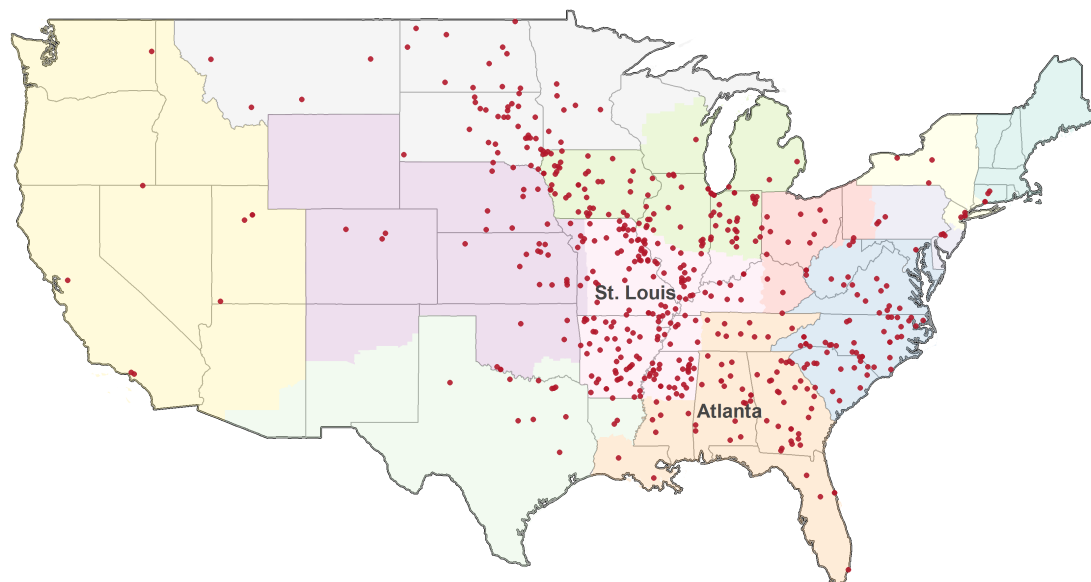


Figure 9. Permanent failures due to Caldwell collapse and the Federal Reserve Structure as of the end of 1930.

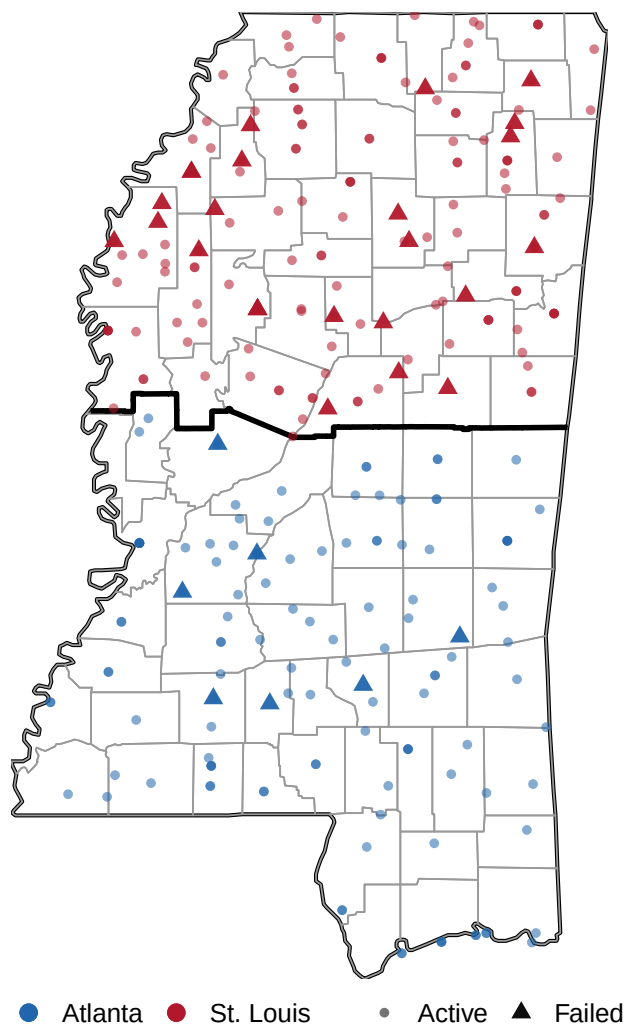
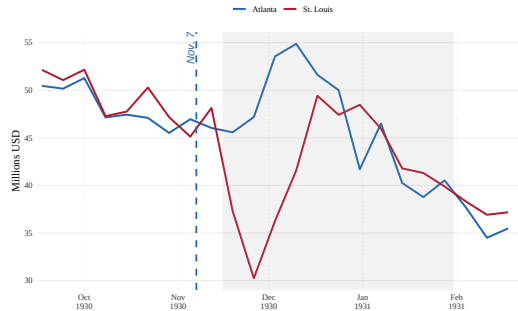
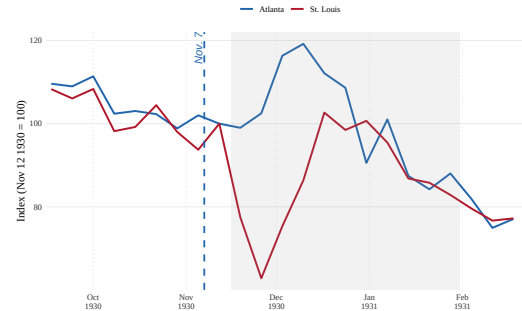


Figure 10. Permanent Failures in Mississippi during the Caldwell collapse, by district.

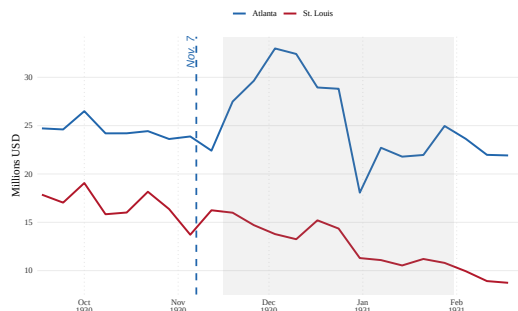
7 Figures



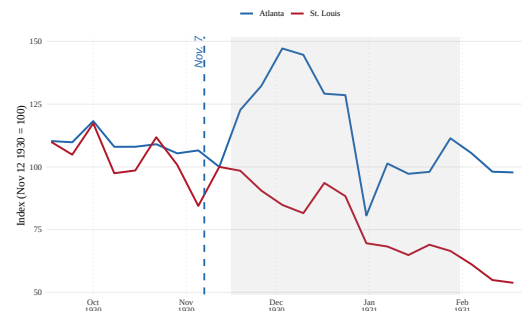
(a) Total Credit (Levels)



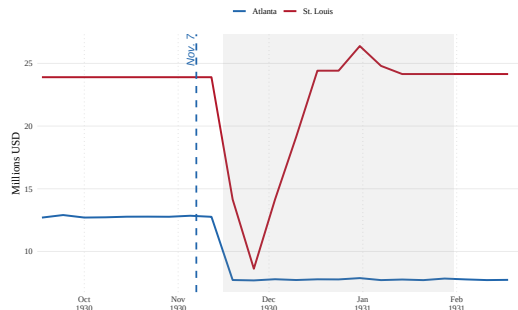
(b) Total Credit (Index)



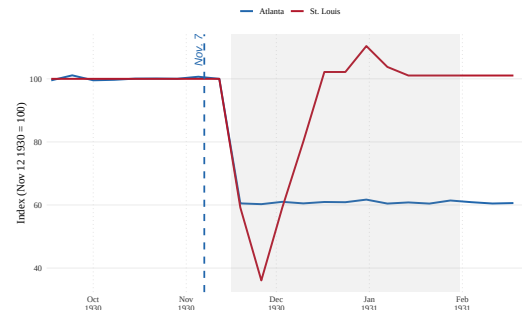
(c) Discounted Bills (Levels)



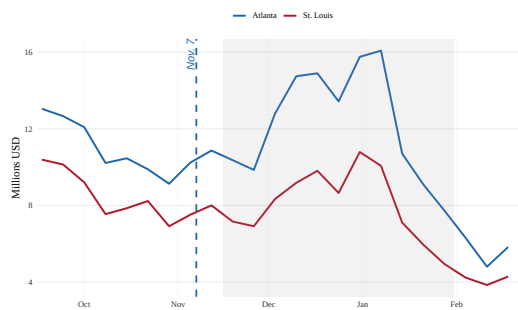
(d) Discounted Bills (Index)



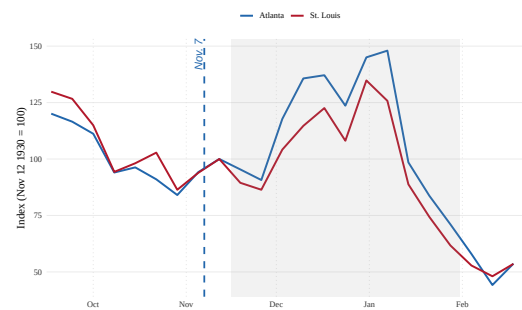
(e) Government Securities (Levels)



(f) Government Securities (Index)



(g) Bills Bought (Levels)



(h) Bills Bought (Index)

Figure 11. Credit Provision in the St. Louis and Atlanta Districts. Each row compares the levels on the left and an index (base Nov 12, 1930) on the right. Shaded area marks the Caldwell-collapse related period. Atlanta in blue and St. Louis in red.

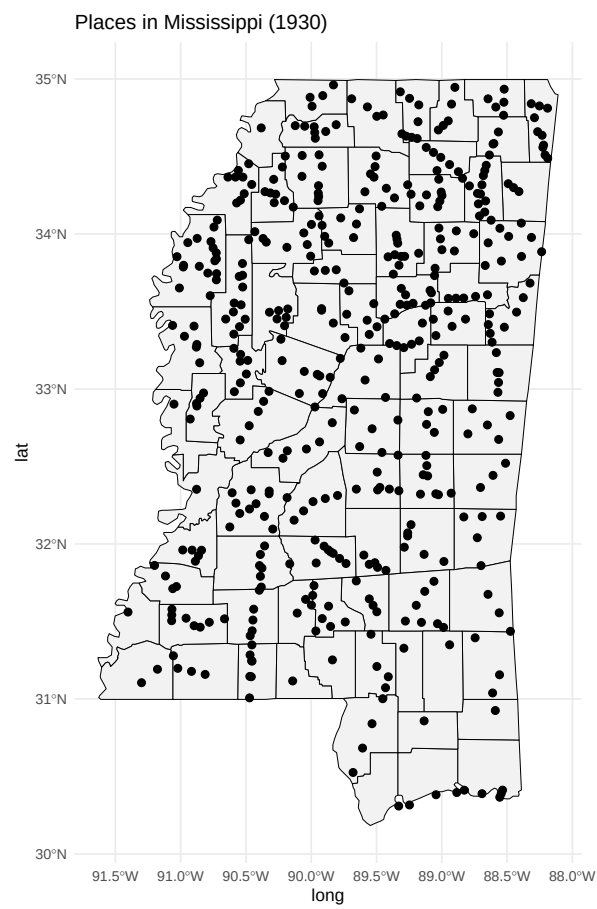


Figure 12. Sub-county Identified Places in Mississippi, 1930. Source: Decennial Census and CPP data.

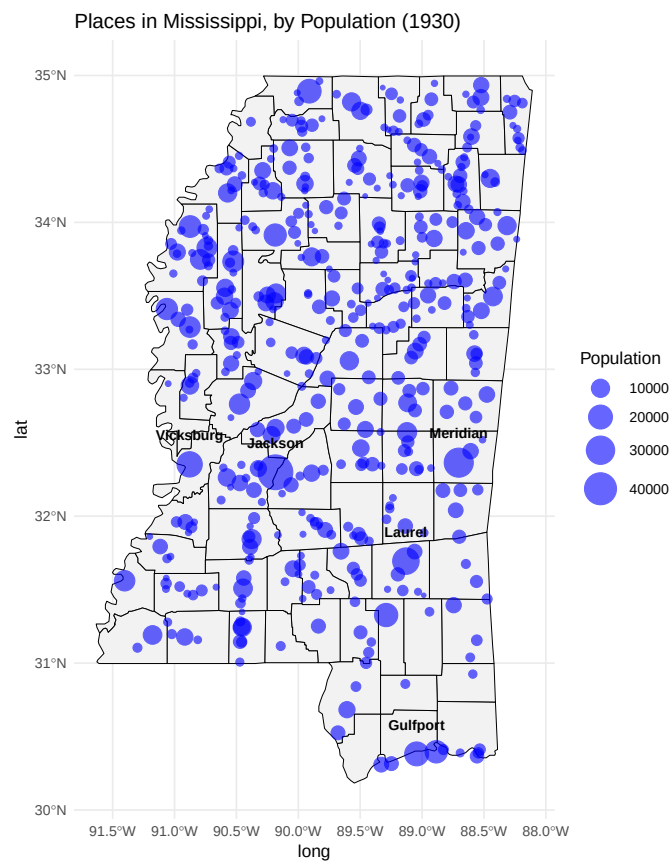


Figure 13. Places in Mississippi by Population, 1930. Source: Decennial Census and CPP data.

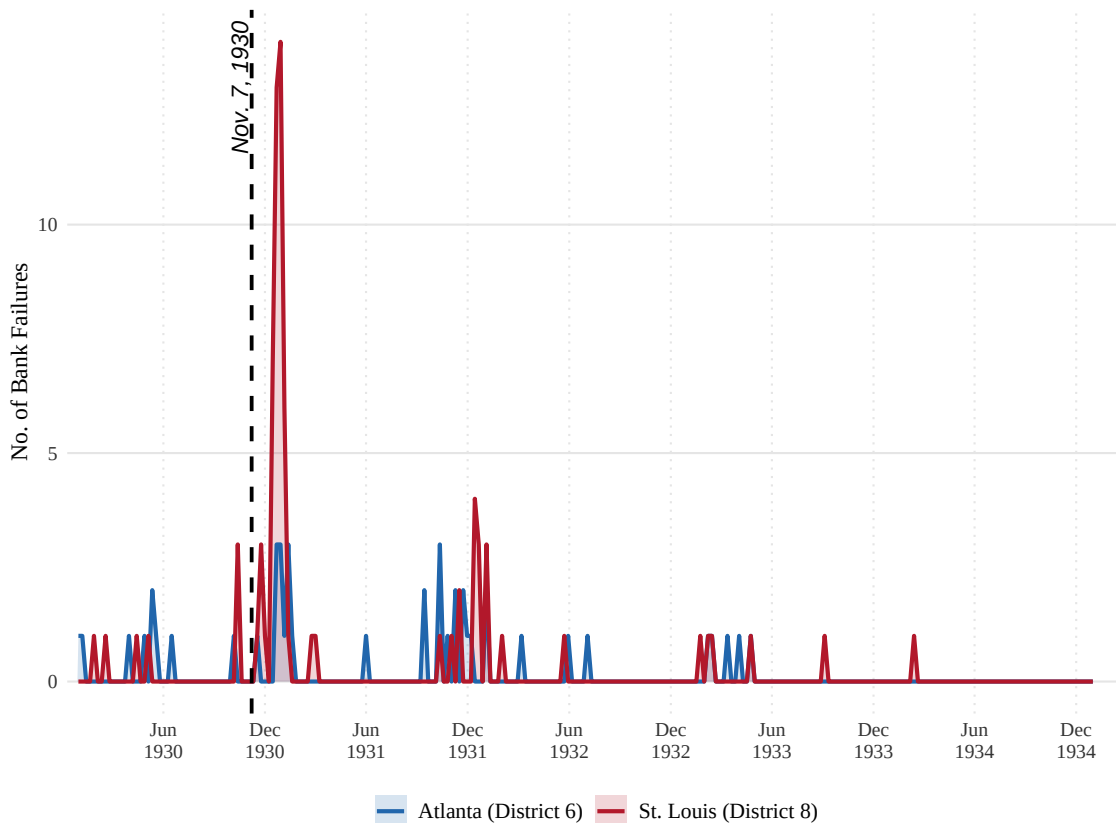


Figure 14. Weekly Bank Failures in Mississippi by Federal Reserve District. Source: Rand McNally Banker's Directory, several years.

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Number under Name of Bank is the New Transit Number given to each bank in U. S. exclusively by The Rand-McNally Bankers' Directory, under the authority of The American Bankers Ass'n.

MISSISSIPPI—Continued										Non-Bank Towns with Nearest Banking Point (Indicated Across.)									
										LAWYERS, LAWS (indicated in back of this volume. For interest rates, holidays, etc., see Laws.)									
										PRINCIPAL CORRESPONDENTS.									
										REVENUES									
										PAID-UP CAPITAL.									
										SURPLUS AND PROFITS.									
										DEPOSITS.									
										OTHER LIABILITIES.									
										Loans and Discounts.									
										Real Estate.									
										MORTGAGE LENDING.									
										Case & Exchange.									
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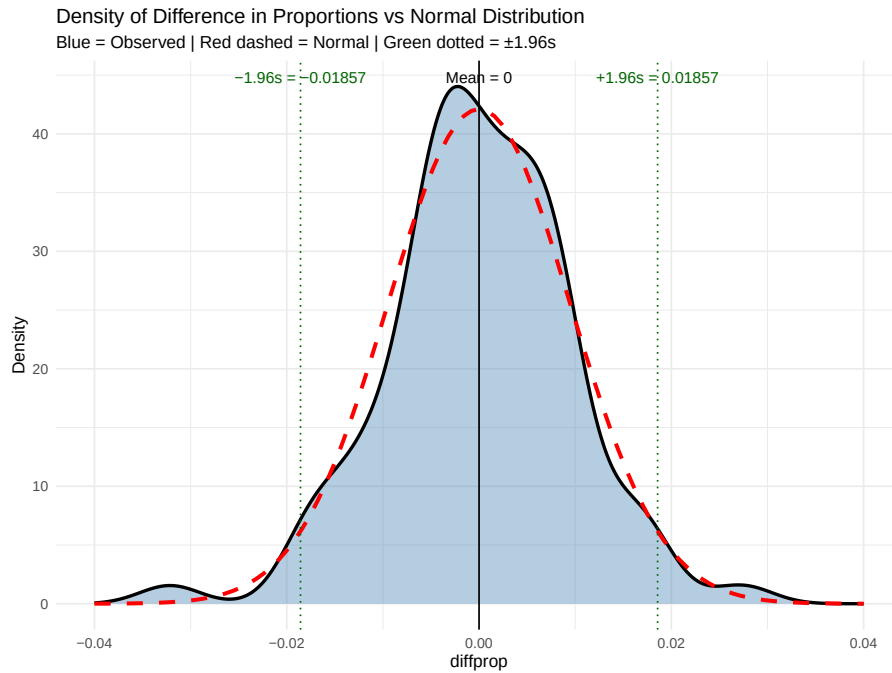


Figure 16. Density of Difference in Proportions for 1930. This figure plots the distribution of the difference in proportions of each county's population in the complete census versus the geo-referenced locations. The differences distribute very close to a standard normal.

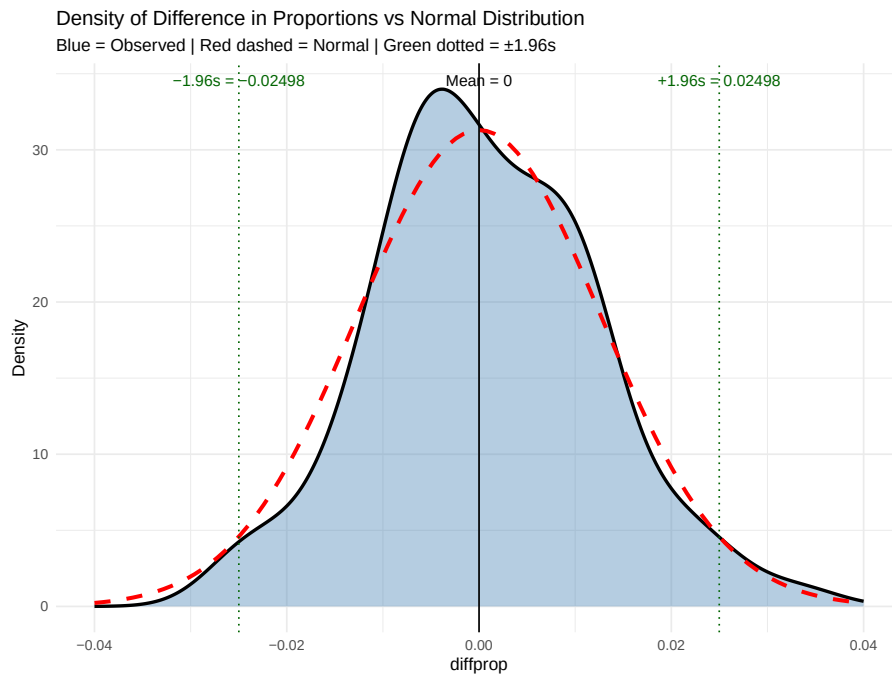


Figure 17. Density of Difference in Proportions for 1940. This figure plots the distribution of the difference in proportions of each county's population in the complete census versus the geo-referenced locations. The differences distribute very close to a standard normal.

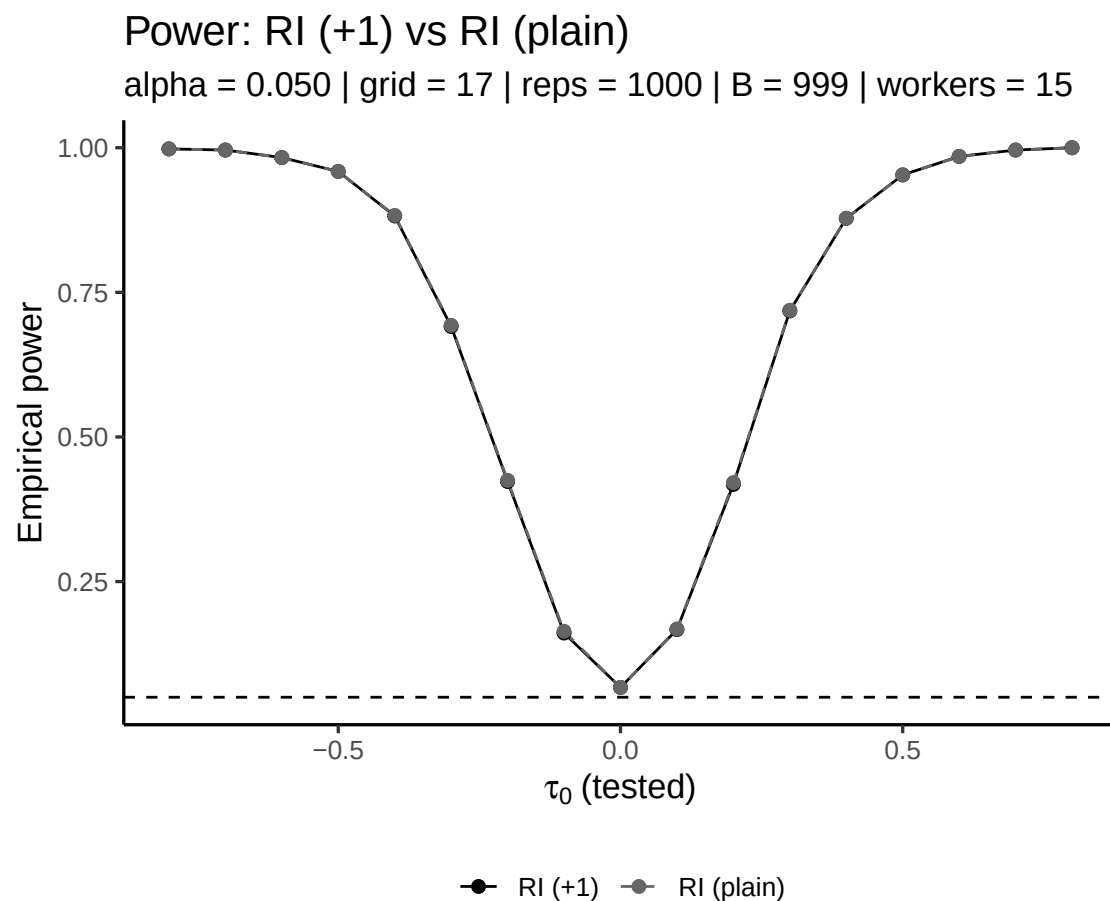


Figure 18. Empirical power curve of the cluster-robust randomization test. Nominal size is 0.05, 1000 repetitions and 999 permutations were implemented. The null tested is in the x-axis. Source: Own elaboration.

8 Additional details

Table 22. Rand McNally Directory Categories: Hancock County Bank Example, Jan 1930

Category	Definition	Amount
Liabilities		
Paid-Up Capital	Actual shareholder capital paid in	\$60,000
Surplus and Profits	Retained earnings and undivided profits	\$135,890
Deposits	Total deposits (demand, time, savings)	\$2,004,110
Other Liabilities	Borrowings, rediscounts, or obligations	\$120,000
Resources (Assets)		
Loans and Discounts	Loans to customers, discounted notes	\$1,314,220
Bonds and Securities	Government and other investment securities	\$639,990
Miscellaneous	Premises, fixtures, and other assets	\$130,330
Cash & Exchange, Due from Banks	Vault cash, correspondent balances, items in clearing	\$235,460
Total Liabilities		\$2,320,000
Total Resources		\$2,320,000